

Psyc 231 Introduction to Abnormal Psychology Part 2

Numbered Transcript

PSYC 231 • Dr. Robert O. Pihl

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Lecture 1: A Rise in Depression

At a glance — A rise in depression.

- 1.1 Hi, this is Abnormal Psychology 2 because there was an Abnormal Psychology 1. So because this is Abnormal Psychology 2, I've got to just give you a brief summary of Abnormal Psychology 1. And in that, it was basically three things. We talked about definitions of abnormality, methods of assessment and ideological theories. And we found them all wanting.
- 1.2 In terms of definitions of abnormality, we did not discriminate—I'm using a collective "we", but that's the way it goes— we didn't discriminate between mental illness, mental health, and abnormality. Some people do, but they often severely overlap. And concerning definitions, the one thing we found is that if you ascribe to the current nomenclature, the psychiatric nomenclature, which is found in the Diagnostic and Statistical Manual of the American Psychiatric Association, there are 543 mental disorders. So that's right now, the cumulative number of disorders that one could diagnose.
- 1.3 The DSM was first developed in 1952 and it had 128 disorders. So you go from 128 in 1952, to 543 in 2022. That's my definition of inflation. That's any definition of inflation. And I will guarantee you that there will be a DSM 6 and a DSM 7 in your lifetime, and the number of mental illnesses will grow accordingly. So we're going to push a thousand mental illnesses, which means we're all going to be nuts. Probably are already.
- 1.4 But the other point that was made was that definitions are absolutely critical because they're the first stage in any kind of scientific advancement. You have to define something before you can even endeavor to predict it or to control it. So we have to have good definitions. And the one point that I'll be making through the next two days of lectures is that we have—
- 1.5 We're going to talk about depression. And I'll give you a bottom line already. And that is: there are multitudes of depressions with differential etiologies, most of which we don't know. But we treat depression as one ubiquitous thing. And it's not. It's a separate series of disorders that have many different ideological causes. And if we're going to deal with them, we have to have differential treatments, and we don't.
- 1.6 We talked about methods of assessment and said that the problems with methods of assessment is that even though this Diagnostic and Statistical Manual sees mental illness as primarily being biological, there are no biological measures in the definitions of mental disorders, except for just a couple out of these 543. So the primary method of assessment is the clinical interview. It's just simply you talking to a psychologist or a psychiatrist or your GP and they determining that you have a disorder.

- 1.7 And the problem with the interview is it's easily manipulatable, just rife with bias. And so there's some people who are really good at it and for reasons that they often even can't explain. I knew one neurologist who was superb at picking tumors. This was actually before imaging, that's how old I am. That he couldn't explain on what behavioral signals he was deciding that this was a parietal lobe tumor or this was a frontal lobe tumor and exactly where it was. But he was good at it. But it was an art form and that's important to know.
- 1.8 We've increasingly, in research, we're in the brain because we believe that mental illness is a brain disorder. And so a tremendous amount of money and research is in that area, but it has not translated to either definition or treatment. So methods of assessment are still rather primitive. Consequently, we have a lot of ideological theories, and that tends to go when you have ambiguity. When you have ambiguity in anything, people have explanations, and most of them are wrong, and you don't know which ones are right. And that's kind of the status that one is stuck with.
- 1.9 So that's a brief summary of Abnormal 1. And the conclusions were that names, definitions, are bestowed, not found. And that's critical because we think schizophrenia is a thing, but it's a word. It's a word used to describe a series of behaviors. It doesn't explain those behaviors. That's another issue.
- 1.10 Diagnostic tools are riddled with imprecision. And that's, as I said, they're very prone to bias and they're easily manipulable. We've done a number of studies where we've just completely tricked shrinks so that they draw the wrong conclusions and they prescribe the wrong drugs. Theories and speculations proliferate. And because of this, current treatment in the area of mental illness is often just trial-and-error.
- 1.11 And there are some people who are really good at treating disorders and they often have very fixed ideas on why they're really good at it. But their explanations could be faulted. But they're still good therapists. That's why I say it's more of an art than a science.
- 1.12 So this is a series of graphs that, I'm sorry, that you probably can't see. But what they show that the lifetime risk, if you have a mental illness, I can't even see it, it grows if you have a mental disorder so that, if you reach the age of 100, we'll try for that, and you're a woman, your odds are about 85% that you will be diagnosable for a mental disorder. And if you're a man, it's about 75, something like that. So it's something to look forward to as you age.
- 1.13 And this is another slide that, again, I will try to point out and tell you what it means. And that is that the cumulative incidence of psychotropic medications in your lifetime, if you're a woman, it's again about 79% that you will have received an antidepressant at some time in your life, or a little bit less, 60-something percent that you will have received an anxiolytic. All of the other drugs are kind of low, but it's antidepressants

and anxiolytics that are prescribed like candy and that which eventually—and we'll go into the psychological dynamics of in fact prescribing these drugs.

- 1.14 So, that you might be able to see. And it's graphs that show current years lived with disability by age and sex. And the bright colors are females, the darker colors are males. And what it simply shows is that it's individuals in your mid-age range that mental illness is most likely to affect you. Whereas the other disorders, these are tumors, heart disease is one of them. And just other diseases such as diabetes and even unintentional injuries. The only ones that are spread evenly over your lifetime are mental illness and unintentional injuries.
- 1.15 Other major illnesses all skew to, as you get older, you become more susceptible to developing these disorders. This graph is designed to show you where we're going in terms of today's lecture, and that is that these bars show the incidence of various disorders at ages—what we're interested in is 15 to 19 and 20 to 24. And if you're in that age range, and you look like it, between 20 and 24, maybe I should have gone older. What it shows is that the major disorders are depression, the red bars, and drug abuse, these yellow bars, and anxiety also.
- 1.16 So it's anxiety, depression, and drug abuse, drug addiction. That's major problems for individuals in your age range. But problems persist that this is a British survey of 50,000 adults and it shows levels of distress from 2020 to 2022. And again, the whole point of it is, is that there's a consistency across time of these disorders. So mental illness has been relatively consistent. A slight growth in severe disorders, but still ending up just slightly below 10% of the British population. So these disorders in this British survey were highest in individuals between 18 and 24, which again might capture some of you. And that's all that shows.
- 1.17 I did want to show you something about what happened during COVID. And if you get into this course lecture when it's actually produced and you see this slide, you'll see that what happened during COVID is treatment for PTSD, schizophrenia, bipolar, depression, and anxiety. And the graph actually shows you the relative amount of individuals that were being treated for those disorders. It just plummeted and stopped existing.
- 1.18 But it was quickly replaced by Telehealth. And Telehealth actually led to the situation where, presently, the total number of individuals being treated is increasing dramatically because people are going back to their in-office visits and they're also using a lot of Telehealth. So you've added a new dimension, a dimension that was by itself during COVID but then they too got added together.
- 1.19 So the state of mental health in America 2023, which is what mostly we'll talk about, although the years will change because years mean nothing to me now. Currently 20% of U.S. individuals or 50 million people are experiencing a mental illness. In the U.S., 50 million, and they all vote. What's interesting to note is that 28 million are never

treated. So approximately half, slightly more than half, are diagnosable but are not treated.

- 1.20 15% of adults have a substance use disorder. 4.8% of adults hold serious thoughts of suicide, and 16% of youth have a depressive episode. 60% of young individuals are not treated, and 44% report persistent feelings of sadness. So that leads us to Abnormal 2, which is what we're going to talk about. And the topics are depression, suicide, and self-harm; it's not a how-to-do-it course. And addiction and substance abuse.
- 1.21 And I'll be discussing these within a theoretical framework. And that theoretical framework considers family history, genetic factors, because there isn't a disorder in which genetic factors do not play a role. In some it's more significant than others; and the role, you have to understand, is always interactive. There are very, very few disorders that you can say are truly genetic. But all disorders are genetic, if you can understand that twist of verbiage.
- 1.22 So we look at genetic factors, family history, and when we look at genetic factors, we're going to look at biochemical factors and how the, the structural aspects of brain. Then we look at how temperament is formed. And temperament can be formed entirely opposite if in fact you are prone to be what we call externalizing, which is an increased likelihood of being antisocial, or if you are kind of internalizing, where you have an increased likelihood of anxiety and depression. And these are totally separate problems with separate pathways.
- 1.23 So we're going to talk about depression first. And I put up a list of names. I'm not going to read them, but there's what, I don't know, 50, 60 names. We go from Elijah to King Saul to Lady Gaga to Bruce Springsteen. These are all people who somewhere have stated or we know to have been diagnosable as depressive or as bipolar. And while that list is up, I'm going to read you something, and it's from a student.
- 1.24 Dear Dr. Pihl, I'm a student in your abnormal psychology class. You've often read us third-person accounts of case histories of people who suffered from various psychological disorders that we're studying. I think this is important as it's easy for us to sometimes forget that psychology is fundamentally about people and that those who suffer from these disorders, as well as around them, and are profoundly affected.
- 1.25 We read many studies, memorize lists of symptoms, prevalence rates and theories, but rarely do we get a personalized account of what it feels like to have a mental disorder. I, on the other hand, know firsthand what it's like. I was diagnosed two years ago as having major depressive disorder. It is only recently, I've begun to feel like my old self. I would like you to read my case to the class—you're the class—because I want people to appreciate that I'm not a statistic in a research article, but a person that you couldn't pick out of a crowd.
- 1.26 First of all, it should be known that I had a very happy childhood. I was not abused, neglected, or unduly punished. I strive for perfection, although I can honestly say that

my parents didn't demand it of me. I am part of a typical nuclear family, mom, dad, sister, and dog, living peacefully in a middle-class neighborhood. I was in a wonderful relationship, had many close friends upon whom I could depend. These are all risk factors I have read about in the class and none apply to me.

- 1.27** Things started going downhill in my second year at McGill. That's not unusual. In retrospect, if you can trace a beginning to these things, which I don't know if you can, then it was probably around January, although it wasn't until late March that I realized that my problem exceeded what I thought was normal and decided to investigate. Ironically, what brought me to the doctor in the first place was that I thought I might have mono. For a long time I had no energy, and getting through the day was a struggle.
- 1.28** While I used to think that napping was a waste of time, I now couldn't get through the day without one. Yet despite my constant state of exhaustion, I couldn't fall asleep at night. And I would constantly wake up at four or five in the morning and would be unable to fall back to sleep. Sure, I'd been irritable, had no desire to see friends or to do anything social, but I somehow didn't see it at the time. I had no appetite for food either. Every time I thought I felt hungry, the thought of putting food in my mouth made me feel sick.
- 1.29** I rarely saw any of my roommates, spent most of my time locked in my room, trying to work through most of the time. I couldn't get my mind to focus. I would try to get through the day carrying out normal routine, but I always had this lump in my throat. I felt like I could burst into tears at any moment for any reason at all. About once a day, I did. When I finally saw my doctor, I explained why I thought I must be sick and started to cry uncontrollably.
- 1.30** She pulled out a brochure on depression, showed me a list of symptoms that described everything I had been feeling. We talked a bit longer, and she told me that she suspected that I might be suffering from depression, but wanted me to go for full blood work just to rule things out. I've never wanted to have an illness so much in my life. Somehow having some crazy virus seemed better than a medical disorder. To my dismay, I was perfectly healthy from a medical point of view. This is when the real battle began.
- 1.31** I'll spare you the details of the next two years, but I will say that it's been extremely difficult and I still don't know what caused this to happen, although I've since learned that there's a history of depression in my family. I have been through various courses of psychotherapy and a slew of different medications. It has taken what sometimes seems like forever, but I can finally feel like the flame inside of me has been rekindled. For a long time, feeling neutral was as good as it got, but I now feel generally happy at times.
- 1.32** In the summer, if my mom would see that I was in a good mood, she would make it a point of telling me, because it was not long ago that I wondered if I would ever be

happy again. I want people to know what I went through because I strongly believe that if I had known more about clinical depression, I might have been better able to see what was going on inside of me and been able to catch myself before I fell so deep.

- 1.33** And I want my classmates, when they read all the latest stats on depression, to try to cram them into their brain for the exam, that this and all other disorders we study can affect you, someone you know, someone you love, or even yourself. A very poignant statement which adequately listed the depression, the symptoms of your kind of run-of-the-day depression.
- 1.34** So let's go over a little history of depression. Says, quote, I didn't get the theme here, "Man that is born of woman is of a few days and full of trouble. He cometh forth like a flower, and is cut down. He fleeth also as a shadow and continueth not. For there is hope of a tree if it is cut down, that it will sprout again. But man dieth and wasteth away, and where is he?" And it's very dysphoric. And it kind of paints a picture of ongoing depression that one is all going to face as we face the struggle, the struggle of life.
- 1.35** The Bible also has cases of depression. King Saul was depressed and was treated by David with harp music and quote, what amounted to good vibes, but just discussion and relaxation. Then along came the Hellenic ages where depression was explained in terms of the presence of the relative levels of humors in the body, with the highest level being, if you're depressed, being black bile secreted by the gallbladder and the intestine.
- 1.36** So that Hippocratical theory of depression was carried on into Middle Ages, where possession by devils also became common, and into the Renaissance period where the first book was written on depression by that guy there. His name is Robert Burton. He was a British cleric and he wrote the book in 1621 called the Anatomy of Melancholia. And in the Anatomy of Melancholia, he talked about etiological factors and treatment. For him, treatment involved bloodletting, which is a hangover from kind of the Middle Ages.
- 1.37** And he also talked about talking with one's friend and confessing one's grief to a friend. That in itself is probably a pretty good definition of psychotherapy, which was being espoused in 1621. But here are some of the causes that Burton wrote about. First of all, he had supernatural causes. So you can't get rid of the Middle Ages. So God, devil, and witches were causes for depression. But then he said there are other causes. In general, one's parents can affect depression; temperament, old age, education, terrors, imprisoned, poverty.
- 1.38** And he said then within each person, your humors, which is Hippocrates in the 1600s, hot brain, excess venery. Does anybody know what venery is? I had to look it up, too. It's too much sexuality. But in the 1600s, you called it venery, which sounds like a song. So the problems of the bowel could produce depression. Distemper of the liver, bad

diet, coarse bread, unclean water, sharp sauces, constipation, the evil of constipation. Thick air, excessive exercise, passions, irascible sorrow, fear, shame, disgrace, envy, malice, hatred.

- 1.39** It's kind of all-compassing. Just about anything can in fact cause depression. That was in 1621. This is data as old because it was a survey done from 1872 to 2003. And it just shows you how interested we are in depression because, during that time period, there were 91,000 articles written on depression, 73,000 on anxiety, 4,000 on happiness, and 1,000 on joy. Kind of shows you where we're focused.
- 1.40** Now, again, that's 2003, and I included that because you can go home and go to your ChatGPT, whatever it is, and you can produce the same data in about five minutes. I didn't want to do that because some poor sod had to pick all these out at some time, and they deserve credit.
- 1.41** What's interesting, speaking about happiness, is compared to depressed people, happy people are less self-focused, less hostile and abusive, and less susceptible to disease. Also of interest relative to happiness is, Gallup has just published their World Happiness Report. And the Nordic countries are in the top 10. Finland's number one, Denmark's number two. Sweden and Norway are between that and the top 10. Canada was number 15, USA number 23. But here's the kicker. For those of you under 30, Canada ranks 58th and the USA ranks 62.
- 1.42** So it's the unhappiness of the young that are affecting the total score of Canada and the US. And what is unusual is, that is really atypical. That's not the rest of the world. That's us. Also, what they found is that, for the US, unmarried individuals are sadder than married individuals, that teenage girls are significantly sadder than teenage boys. And if you are politically bent, then you're really bent anyway; liberals are sadder than conservatives, and that's been true for decades. And liberals are also more likely to report having been diagnosed with a mental illness.
- 1.43** Now, we could probably sit and look at our navels and figure out what that means, but it's not worthwhile. It's too rancorous. So back to depression. But there's a kernel there that you have to grasp because we're going to have to talk about it. And that is the mood of individuals your age, because it's an outlier.
- 1.44** National Center for Health Statistics. Antidepressant use is up 400%. Don't pay attention to that because I don't know where it comes from, the start date is. But we do know that it's up 64% in young teens and young adults since 2020. So four years ago, the use of antidepressants is up 64%. It involves 1 in 10 Americans, 23% of women in their 40s or 50s. 14% are white, 4% are nonwhite, and less than a third have seen a mental health professional.
- 1.45** This is an interesting study and it relates to where all these drugs are coming from. There are three scenarios here. Let me tell you what the study was, first of all. What they did is they wanted to check the validity of prescriptions that GPs were giving for

antidepressants. So what they did is, they selected, they got a number of GPs to agree to let them take patients that they've seen and do complete psychiatric psychological assessments.

- 1.46** So what you have to know is that most prescriptions for psychotropic drugs are written by GPs, general practitioners, not by shrinks or psychologists, because psychologists in some states can write prescriptions too. So you have these people who are just, they see their GP and they do or do not get a prescription. They're seen again by a psychiatrist and psychologist who do another diagnosis. So what you can determine is whether you have false positives or false negatives.
- 1.47** What's a false positive? That's where you say somebody's depressed and they're not. And a false negative is where you say they're not depressed, but they are. So we're going to view our shrink and our psychologist as being a gold standard. They're not, but we're taking it that way. And so we're going to be able to draw a chart where the original GP was right, saying that a person was depressed when they were depressed, and when they're false negatives, when they're wrong, false positives when they're wrong, where they say a person is depressed but is not depressed, and they've written a script for them.
- 1.48** And we're going to be able to look at false negatives, where I say a person is not depressed when they are and should be treated. But what this, if you can see the graph shows, is the proportion, given the assumption of the number, percentage of depressives that walk through the door, and if we assume that that's 20% of people walking through the GP's door are depressed, then it turns out that he's wrong or she's wrong 25% of the time. 25% of the time, against the gold standard, he's going to say that person's depressed or not depressed when she's wrong. And that's probably the frequency that exists or close to it in reality.
- 1.49** Now, subsequently, companies like Pfizer have produced little checklists for physicians, 10 items where they can—I think I'll show it to you later—where they can in fact do a diagnosis, but most still don't. So what's the prevalence? In a community sample, it's, in a lifetime, 10 to 25% for women. So that means, ladies, up to, your chance is about a quarter that you're going to be depressed in your lifetime. For males, it's basically half of that, 5 to 12%. At any point in time, it's around 10%. It's 5 to 9% for women and 2 to 3% for males. And again, 80% go untreated.
- 1.50** What's the consequences and costs of depression? Well, there's suicide, and we know that 70% of individuals who commit suicide were clinically depressed. The risk for suicide if you are depressed or have a mood disorder is 15%, which is 15 times higher than the general population. In terms of how that affects disability, we look at what's called DALYs, D-A-L-Y-S, which is disability-adjusted life years. In 2020 for unipolar depression, it was second after heart disease, but the first cause for women.

- 1.51** Cause of depression in Canada, for women, for everyone, is \$27 billion a year. In the US, 326 billion. This figure just simply shows that, in terms of days spent in bed, depression is second to heart disease as a medical condition. But here's something we start getting interested in, because statistics have been relatively consistent, but they're changing. And what this figure shows is, for young women in particular, there's been, from 17 different countries, there has been a dramatic increase in suicide rate. And this figure looks at percent change since 2003.
- 1.52** And the highest change are girls in their teens, followed by women in their 20s and women in their 30s. There's a gradual increase. A survey was done of 17,232 high school students in the US, and teen girls scored worse than boys on almost all measures. They were 57% greater at feeling sadness or hopelessness, 30% had considered suicide, and 18% experienced sexual violence in the past year. 70% reporting non-heterosexual identity reported sadness, hopelessness, and 22% had attempted suicide.
- 1.53** It is that statistic in particular that has had a dramatic effect on how society thinks and attitudes in early education. Because the theory has been that its lack of acceptance, that is in fact the basis for the statistic. And that's a legitimate question to ask—is that true? I'm sure it's partially true, but how much does it explain? And that's something one really needs to ponder. And we're not. What it's done is produced a great deal of division, tension, and general societal issues that we're not going to discuss.
- 1.54** Anyway, for one year, I think this was 2022, major depressive episode for youth, it affected 2.3 million in the US which was basically 10% of all youth. And these were individuals who had severe impairment. Another almost 1% didn't have severe impairment. This is a fascinating study which, which makes a really important point. These are trajectories, and a trajectory is where you follow a disorder over time because nothing's stable. You know, if you're depressed on day one, year two, you may not be depressed. We'll see that in a minute.
- 1.55** And it's true of all kinds of disorders. But in this case there were, if you can see the curves, they look almost identical. And they are trajectories of depression in two different samples, one sample in the United States and B is a sample in Great Britain. And you get data that looks exactly the same. So when you get reliability like that, using different measures, both looking at depression, though, and the trajectories are four different trajectories in samples of kids, large numbers of kids, it tells you a lot.
- 1.56** And you see that there's kind of a parabolic group that at age 10 start showing signs of depression and they peak around 13 and then drop. And then there's another group that, when you first see them at age 10, they're high, but untreated, they're fine after age 14, 15, 16. If you're a psychologist and you're seeing kids, or a psychiatrist, and you see a 10-year-old that comes in and they look depressed, you don't know which group they're in. This study didn't actually discriminate about who fit each group, other than they had these general measures of depression.

- 1.57 But then there is a group which is called an increasing group, where the curve goes up with age, and that's the group that's problematic, and that's the group that you have to worry about. And those are the individuals that you have to deal with. But what it illustrates is that, at one point in time, age 10, you've got a whole bunch of kids who are doing the same measure. And some look very depressed and others don't. But that's not the story because you have to figure out which trajectory they're in, in terms of where they're going to end up.
- 1.58 This is just a line which shows how the only group that is increasing in depression over time are individuals who are 18 to 25. And here's something that I know you can't see because I can't even read it here, but I'll just describe this large amount of English. There's a group called the US Preventive Services Task Force. It's a group of experts paid by the US government. Basically, they are hundreds of different committees that are put together to answer a specific question.
- 1.59 The questions, for example, the one I was going to show you is, should screening devices for depression be used for adults, pregnant and postpartum persons, and older adults? They ask the question. The first thing they say is, is this an important question? And they say, because of suicide and other things, it's a good idea to detect who has a likelihood of developing depression. And then they go through, what are the benefits of early detection? And if you have, if you can early-detect a disorder, can you do anything about it? Can you do anything about it that is meaningful? If you can't, don't detect it. Leave it alone.
- 1.60 So that's basically the question they answer. And they say, yeah, there is things you can do if you detect depression in adults or pregnant individuals. But then they also ask, is there any harm to detecting it? You know, sometimes finding a disorder so changes a person's way of living that you ruin their life. And sometimes it's better not to know, particularly if you can't do a damn thing about it. So they asked that question, and they came up with, well, yeah, we can do things, so—
- 1.61 And another harm they deal with are drugs. Is a person going to get prescribed drugs that are going to have major side effects? You're always doing a balancing act between — I mean, imagine what a surgeon goes through every day in terms of, particularly a brain surgeon, of whether to muck or not to muck. And so they come to the general assessment and they conclude that, yeah, it's a good idea to look at depression and to analyze it and to test it.
- 1.62 So here I ask the question, is depression a chronic episodic disorder? And I put the question mark there. Previously, I didn't have a question mark, but then I read a paper where people argued that, no, it's not inevitable that it's a repeat disorder, because if you've had one episode, the chance is only 50-60% that you're going to have another. But first of all, the average duration of an episode for depression is between six and nine months if untreated. Thus a natural remission can occur.

- 1.63** I used to tell my graduate students in clinical psychology that if you have a depressed patient, make sure you tell them that therapy will take about six months, because if you don't help them, time will, and you can take credit. Not fair. It's business. Anyway, that's facetious. The general length of a depressive episode is hard to predict. Two weeks to several years. But here's the important thing, and this is why you have to worry about it being chronic and repetitive.
- 1.64** If you've had one episode, there's a 50 to 60% chance that you'll have a second. If you've had two episodes, there's a 70% chance you'll have a third. And if you've had three episodes, there's a 90% chance of having a fourth. So let's look at some symptoms. This is basically a factor analysis of the Beck Depression Inventory. So there are four symptom clusters—physical, affective, cognitive, and motivational.
- 1.65** Physical symptoms, you saw that in the case I read, primarily a sleeping problem and an appetite problem. So you get people who can't sleep or sleep too much, can't eat at all or eat too much. Generally, depressives tend to put on weight, so they eat too much. And sex drive, I say that to students, and they say, what? But mostly it's either too much or too little. The same kind of thing. What's interesting about these physical symptoms is a lot of it's made out of them, but they correlate the least with the other three, affective, cognitive, and motivational.
- 1.66** In fact, it's affective or emotional symptoms that are the essence of depression. One feels miserable, hopeless, blue, sad, lonely, unhappy, guilty. It's like you're walking around with a black cloud that follows you everywhere. And it's how you see the world and you think how the world sees you. Cognitively, you have negative expectations; you can't make decisions; you're very indecisive; you have trouble concentrating, and you will actually cognitively distort things.
- 1.67** Later on, we'll see, what's interesting is, you see a lot of what's wrong around you. If I take you into a scenario where I want you to watch two people interacting with each other, and they're actors following a script, and one of them is purposefully sticking a needle in the other with subtle negative statements, the depressive sees that right away. If you're not depressed, you don't.
- 1.68** So there's an aspect that often the depressive is closer to reality than the rest of us because they see the shit, and we don't, because we're so concerned with ourselves that, if we're talking with somebody else or we're into a conversation, it's, how do I fit in that? And what can I say? And you know, you're coming from your position, you're not seeing what's going on. Anyway, so there is this cognitive distortion, but it's not always negative. It's negative in terms of you seeing what is negative.
- 1.69** Most of us survive because we have rose-colored tinted glasses, and so we kind of put a good spin on things. I mean, how many of you think about the end of the world on a daily basis? It's going to happen, actually billions of years from now anyway. Motivationally, person could be suicidal. They certainly underestimate positive

reinforcement, overestimate negative feedback, and don't believe that they can affect things, which is low self-efficacy.

- 1.70** So a major depressive disorder, this is the diagnosis. You have to have five or more of the following symptoms and they have to be present during the same two-week period and represent a change from functioning. And one of them is either number one or number two. Number one is a depressed mood most of the day, nearly every day, as indicated by subjective reports. Feel sad, empty, hopeless, etc. Appears tearful or markedly diminished interest or pleasure in all or almost all activities.
- 1.71** Then significant weight loss when not dieting or weight gain, insomnia or hypersomnia nearly every day, psychomotor agitation or retardation, fatigue or loss of energy, feelings of worthlessness or excess, inappropriate guilt, diminished ability to think or concentrate, indecisiveness, recurrent thoughts of death, suicide; and then the symptoms cause significant distress, and the episode is not attributable to psychological effects.
- 1.72** So during a one-week period—because the final criterion was not a manic episode—so during a one-week period, at least three of the following: inflated self-esteem, decreased need for sleep, more talkative, flight of ideas, distractible, increase in goal-directed activity. And what you see in a manic episode is that you go well above your normal high baseline. And if you ever, it's hard to deal with a manic patient because, for many, their manic state is very rewarding. It's also very troublemaking because they get into trouble, they overspend, they do things that we normally inhibit.
- 1.73** I was walking down a street in Montreal with a manic patient. We walked past the bank. He said, "Hey, wait a minute." He went in the bank, and I'm standing on the street, and time is going by, and I finally got tired of waiting after like 20 minutes. I went in the bank, and he's there with the manager trying to float a loan to buy a ski hill. When we had walked past, when we started walking past the bank, a truck drove by, one of those billboard trucks, advertising a ski hill.
- 1.74** He had not thought about a ski hill at all, except the truck went by, "hey, a ski hill!" Into the bank, asking for a loan, trying to negotiate a loan. I mean he was a great bullshit artist. And so it was, he had the guy kind of in his hand. He would never get the loan. But they were working on it.
- 1.75** Kay Jamison, a famous psychologist who wrote a beautiful book called *An Unquiet Mind*, has bipolar. And she so enjoyed her highs that she was really reluctant to take to take lithium. But her lows were so crashing that she had many suicide attempts and finally decided she wanted to live. So she takes lithium. And if you know, or are ever diagnosed, look up her website. It's great material on what to do, how to do it, and very convincing because she's been there.
- 1.76** I'll just skip this. So there are—the one thing that's going on with depression is that there are a number of depressive disorders. The DSM lists 1, 2, 3, 4, 7. There's a

disruptive mood dysregulation disorder, which is basically somebody who has severe temper tantrums. It can even start at a young age, in fact after six. But it's the kind of person, adult who is, who you don't want to be around because you can't determine when they're going to fly off the handle, and they show wild emotional changes.

- 1.77 Then there's major depressive disorder which I read you the symptoms. There's persistent depressive disorder or dysthymia. And the thing about dysthymia is that it's basically major depression lite. So you don't fall into the extremes of depression, but you have some symptoms. Then there's premenstrual dysphoric disorder, which is diagnosable depression one week before menses. Then there's substance medication-induced depressive disorder.
- 1.78 There's also a seasonal pattern of depression disorder, which living in Canada we should pay attention to, because as you know, we in fact evolved from bears and not primates. So we should hibernate periodically. But what's interesting about this seasonal pattern of depression is, though, treatment that works is a bright light. I'm saved. Just here you can buy lights that produce 2,500 lux brightness, which is five times the light in a normal office building, fluorescent light. So it's bright. And what you do, and I had one, Jordan had one; at McGill, we all have them.
- 1.79 And you just use it a couple hours every morning and you can think that maybe that's it that saved you from seasonal pattern disorder. Well again, seasonal pattern disorder is much more likely in women four to one and yeah, so full remission at certain times of year, and it responds to light therapy. We have postpartum depression, peripartum and also postpartum. So symptoms during pregnancy or postpartum. And it affects 9% of women during pregnancy and 7% between birth and 12 months. And it can be severe; it can have psychotic features.
- 1.80 There are a number of cases, some not that far ago, of women who killed their families, suffering from postpartum-onset depression. So this slide just simply shows how important it is to treat perinatal and postnatal, not only because it can be quite consequential, but because it can be consequential also in terms of affecting the attachment between the mother and the child. And that can have long-term effects.
- 1.81 I think one could argue, probably 90% of pathology has its basis in childhood. And many of the disorders don't appear until later in life. But it has to do with from womb on, in terms of how things can impact development and then interact with stressors as they occur in life. And we don't pay enough attention to what can go on in early childhood.
- 1.82 So I present this graph because it shows how there's a dramatic increase in percent clinically depressed that begins at age 15 and how it really is severe for girls, much more so than boys. Now it's important that I point out that this graph is 30, 40 years old. So it precedes handhelds because you'll see that the graph has moved down, but it's pretty much the same, although it's been increased, but the finding has always been there.

- 1.83** So it's not just what's in your pocket. Yeah. So here's that, here's that graph, and that shows, starting around 2011—first of all, the girls are younger, 12 to 15, but we know that menses has become younger and we know that you can see in this graph how it's not just suicide, but self-poisoning, major depressive episodes, and depression itself starts a dramatic up curve.
- 1.84** Now people who produced that last graph argue that this is all because of screen time. And what I'm arguing is that they're partially right, but not totally. And they say that media is linked to mental health issues and suicidal ideation. That is true. It's linked to decreased face-to-face interaction, and that's true. It's linked to cyberbullying, yes. Sleep loss and knowledge about suicidal methods.
- 1.85** So this is data that supports that kind of a view because there are three graphs. And the first graph shows what's happened over years about not hanging out with your friends. And it's for 12th graders, 11th graders, and 8th graders. And you can see there's a dramatic slope. People stopped hanging around with their friends. But you can see these curves start to turn down in the early 90s, 1990s. So it's not just— when was the smartphone invented? When did it become popular? 2012. Isn't that the date typically people point to, or something like that?
- 1.86** But the curves are already coming down. There's significantly less dating. And the last curve is people feeling more likely to feel lonely. That's increasing. Furthermore, that you can't see, it's for those folks at home. But it's an important study, so I'll have to talk about it. It's an annual survey done in South Australia, where they look at grades four to nine on a number of dimensions and what they found. So this is everybody in the school system. So you're talking about between 50 and 70,000 kids are doing a survey.
- 1.87** And what they found is the first curve shows a dramatic drop from 2017 to 2022 in satisfaction, in optimism, downward curve in happiness, a downward curve in cognitive enjoyment, a downward curve in emotional regulation, in perseverance, and increases in worry and sadness. So these are young kids, average age 12. And they're unhappy and they're troubled, and it's a source of concern.
- 1.88** This was a study of 6,595 adolescents where they looked at individuals who had mental health problems if they had internalizing disorders, i.e., depression, externalizing disorders, conduct disorder, for example, or both. And what you can see is time spent on media increases with the development of both of these disorders. So media is a culprit and is involved. And if you want to read statistics about it, the city of Seattle, it's got a big lawsuit going against major providers and they detail all the literature of what's involved.

Lecture 2: Inside Depression

At a glance — Inside depression.

- 2.1 There are many things that can affect depression. So I just read a paper of a five-year longitudinal study with 500-plus individuals where they looked at these people, individuals, over a six-year period. What they studied were their sleep patterns, repetitive thinking, and whether they showed indications of depression, anxiety, or eating disorders. And what they found was because they were looking at things longitudinally, what was important was the level of their sleep problems initially.
- 2.2 And the sleep problems correlated to repetitive negative thinking, which in turn, those two things led to depression. So fundamentally, what they were doing was watching the development of depression in a group of individuals, mostly young individuals. And it turned out that the sleep problems was the most significant predictor. But the studies that I showed you previously, all illustrated that there's this significant, repetitive difference between males and females. People have tried to explain that in many ways.
- 2.3 First, they argue that there's no difference, that there is what is called masked depression. And masked depression is where you find depression, where people are engaged in behaviors which you think are depression, but they're not. Like, men are much more likely to have problems with alcohol. It's said by people who buy into this concept that their drinking is an expression of depression. That's masked depression. But that's not testable. It's just really an idea.
- 2.4 Then a lot of explanations have evolved around cultural and role differences. Women are more likely to be victimized. There are chronic strains that exist in roles, and that's argued to be a factor. Then there's the biological, estrogen et al, the hypothalamic-pituitary-adrenal axis. And dysregulation in brain functioning. Just simply a number of facts. Oral contraceptives and hormone implants may trigger the onset of depression. 20% of women experience depressive symptoms during pregnancy, and 10% develop major depression.
- 2.5 10-15% of new mothers suffer from postpartum depression, usually within two weeks to six months after delivery. And women with previous episodes of depression during other reproductive events, postpartum, premenstrually, are likely to experience depression during menopause. Finally, hormone replacement therapy, especially progesterone, is associated with depressive symptoms in postmenopausal women.
- 2.6 This is just a data which shows oral contraceptive users versus non-users, and it shows, particularly in individuals below the age of 19, that the oral contraceptive users are more likely to score higher in depression. Oral contraceptive users are more likely to reflect crying, hypersomnia, eating problems, tiredness, loss of energy, et cetera.

- 2.7** So this sex difference that exists, there's strong, powerful biological evidence to explain the difference. So let's look at the etiology of major depressive disorders in general. We'll look at the biological and the social, bearing in mind that the disorder itself produces effects. If you're going to study a group of depressives, biologically, they're going to be different because they're depressed. That doesn't mean that the difference caused the depression. What this simply shows is what happens to your body if you're depressed.
- 2.8** First of all, there's a disruption in neurochemistry. The relationship between neurotransmitters that we're interested in can't be discerned. Then the stress system is activated, the pituitary functions, so cortisol is released. The immune system starts functioning and cytokines are released. There are all kinds of problems that occur within the individual that can be mistaken for the effect of depression. It can be mistaken for etiology, rather, and it's not the effect. We have an overall model that you can look at, where biological differences affect psychological vulnerability, affect stressful events.
- 2.9** Then that interacts with activation of stress hormones, problems with interpersonal behaviors, and it all can lead to depression. The problem is, all the arrows are going one way. That's wrong. All the arrows should go two ways because your behavior is going to affect your biological functioning. The end product can be different than a one-way directionality.
- 2.10** What do we know about the biology of depression? We know it runs in families. This data shows the likelihood of depression given whether your parent or both parents are affected or not. You can readily see how the curves are far more predictive when both parents are depressed. This is a forest plot, which basically says the same thing, but it was done with the entire population of Denmark. So you can't say it was a small sample. And it shows that the more inheritance that's passed on, that increased the likelihood of depression, as does this, that just simply looks at father's depression and risk for offspring in over seven million.
- 2.11** The other way that we know it's genetically involved is through twin studies. The green bars are monozygotic twins, same egg, versus dizygotic twins. You can see, for any mood disorder, for particularly bipolar disorder, and for severe depression, you can see an increase of heritability. Severe depression means three or more episodes of depression. But when you're looking at that group of treatment-resistant depressives and that it runs in families, which is what this graph shows, but it shows that depression is the same as schizophrenia and the same heritability, a little bit less than bipolar.
- 2.12** So there might be a generalized heritability factor rather than depressive genes or schizophrenic genes, per se. So a lot of people are out there looking at particular genes, and they're doing gene localization studies. What this simply shows is the number of studies it takes to find a gene loci. For depression, it takes a lot of studies. They're

harder to find. New loci by number of cases. For schizophrenia, you can find a new gene in every 230 cases. For bipolar, you find it in one in every 400 cases. And for depression, one in every 1,500.

- 2.13** Some genes that have been found affect the glutamate system. And glutamate is a major neurotransmitter in the brain, and it's an excitatory neurotransmitter. The standards, serotonin, norepinephrine, have also been determined. But this study showed something that, it shows an interaction. They're predicting depression. And what they found is that individuals that had a variation in this dopamine genotype who had been rejected by their mothers were the only ones to develop depression. It was a study done with a couple of hundred Russian juveniles who were locked up in Siberia.
- 2.14** You can do studies everywhere now. They had good records of whether they were rejected or not, did genotypes, and found this particular relationship. So it was only the ones that had this particular variation that became depressed. Again, there are indications that something's going on earlier than expected. This was a study where they looked at blood biomarkers in a group of individuals, actually a large population. I think it was done in a Scandinavian country where they just do it. They have medical records on everybody, so they can go back and look at, put things together. They did that.
- 2.15** What it shows is from 0.0 on that graph, they were able to go back 30 years and show differences in glucose, triglycerides, cholesterol, and two fats in the bloodstream that were different 10 years or more before the people were diagnosed. These are individuals diagnosed late in life, and they can look back 30 years and see differences that started developing very early.
- 2.16** Biochemically, people have been focusing on the catecholamines, dopamine, serotonin, norepinephrine. It's long been thought that shortages or some difficulty in the functioning of these major neurotransmitters was thought to cause depression. A very early study just simply looked at blood levels of norepinephrine. The gray bars are people with depression, and you can see that they're reduced in norepinephrine. Then the first two are major metabolites of norepinephrine, and then a metabolite of dopamine, and the metabolite of serotonin.
- 2.17** But it was norepinephrine that showed the greatest effect. For a while, norepinephrine alone was seen as a major causative factor. People started buying into a serotonin explanation. That's because we know that serotonin is involved in mood, sleep, appetite, the kinds of things that get affected in the symptoms of depression. It also has an integrating effect in terms of organizing how general functioning occurs. People started looking at abnormalities in tryptophan metabolism. Tryptophan is the major precursor to serotonin, the major protein.
- 2.18** But what also drew attention is, if you create a graph like this, which shows the effect of various drugs on depression, you will see that there is a theme that comes through.

That theme has supported this serotonergic explanation. First of all, reserpine is a drug that was used for hypertension. What it causes or can cause is depression, to the degree that some people who take reserpine have to be hospitalized and treated for depression. But it's a drug effect. And what it does is it reduces serotonin. So one drug effect causing one effect. And then what happens if you give people a serotonin reuptake inhibitor or other drugs that affect serotonin is you produce an antidepressive effect.

- 2.19 What does serotonin do? It blocks the reuptake of a serotonin by a neuron. So there's more serotonin in the synaptic cleft to stimulate the dendrite of the accompanying neuron. So it facilitates neuronal functioning. Don't pay attention to Redux. That was a anti-weight loss drug that functioned when this picture was designed, and it really worked, but it killed a few people. So they decided to pull it off the market. But it did a similar thing, only in a much more dramatic way.
- 2.20 So we got involved in doing tryptophan depletion studies. And that's where you give a person an amino acid mixture in the morning after they follow a particular diet which is relatively free of tryptophan. But this amino acid mixture does not have tryptophan in it, so it doesn't have the precursor to serotonin. And by doing that, we can, within four or five hours, deplete your brain of serotonin rather dramatically. And initially, we started getting findings which showed, yeah, it really looks like the serotonergic theory is the right one, because people kind of became more depressed.
- 2.21 We had them doing, shadowing tasks, where you hear a story in one ear, which you have to pay attention to. And in the other ear, we play stories that are benign or stories that are dysphoric. And we're wondering, can we interfere with what you're supposed to pay attention to? And it turns out we can. We can do that by depleting your tryptophan. So the garbage that you shouldn't be listening to, you do. This study that's here just simply shows, can we get you to so reduce your mood that you might be called depressed after you've taken this depletion mixture? The answer is for some people who are relatives of a depressed person, yeah.
- 2.22 That was taken by many to support this theory that there's a deficiency in serotonin. But it turns out the picture, like everything, is far more complex. We started getting findings with all kinds of groups where you could produce different argumentative effects. And so, the model is partially correct, but it doesn't explain the whole thing. And I think what's going on is that it's true in some subjects, but not in others. These people are the ones that are going to be responsive to a serotonin reuptake inhibitor because it will increase serotonin in their brains.
- 2.23 We know, for example, that if you look at individuals in terms of PET scans and blood flow during sadness, you get these changes where there is less serotonergic activity in the neocortex, the hippocampus, the caudate, and the putamen. But there are also other data out there that looks at brain functioning in general.

- 2.24** This comes from that large-scale Enigma grant, where there are 135 different labs. Is that right? No, 45 labs from 19 countries. They're producing thousands of these models. And so far, they've shown in adolescent depressives that there's lower hippocampal volume, hippocampal and amygdala. There's lower cortical areas of the brain. These are aspects of the brain involved in emotional control.
- 2.25** For adults, they found similar findings. They even found in the last slide differences in the functioning of white matter in the brain. This study just simply looked at anterior cingulate gray matter density, which was reduced. But that reduction, the correlation is with the Beck Depression Inventory. The more you reduce that area of the anterior cingulate, which is the area just back of the frontal lobes, the more likely you are to be depressed. So one can look at differences, neuroimaging. And that slide is just simply a summary of neuroimaging differences between depressives and bipolar patients.
- 2.26** And you can see between the red, that's what's significant. And there are significant differences in a lot of areas, particularly involved in emotionality with major depressive disorder. If you summarize that brain stuff, what you find is that there are differences cortically, frontal lobes, dorsolateral medial frontal lobe, ventrolateral prefrontal lobe. It's pretty much of a frontal lobe phenomena. But the issue is that the cause or the result of depression. Because we know these are areas of the brain that are involved in executive functioning, goal-directed behaviors, planning.
- 2.27** Symptom-wise, they would contribute to cognitive impairment, depressogenic cognitions, suicidality. Subcortically, there are differences in the cingulate, the amygdala, the hippocampus, the caudate, putamen, the thalamus, all of which can affect emotional memory, stress response, regulation of drives, lead to anxiety, vegetative symptoms. If you're walking away with what's really going on, it's pretty much a lot of the brain is just different. It's hard to—when we're looking at these areas, we've covered quite a bit of it.
- 2.28** Things are still in doubt. So why not use AI? And this was the first attempt to use AI. This was done by a lab in Germany, where what they did first is they took all of their data from two labs. It was not just MRI data, but it was biochemistry and behavioral data. Looked at all the data in as many different ways as they could. In fact, they had some millions of numbers of variations that they looked at. They did reliability checks, put it through different schemes. And so they ran this thing in many, many ways.
- 2.29** The problem was, now they're just picking depression, not depression. And their accuracy was only 62%. And they said it's considerably lower than expected. But what was important is they couldn't pick out any single case of depression out of a large sample of depressed and normal individuals. Now, that's very early stages, but I'm sure that'll be redone with larger numbers. See, the point I've been trying to make throughout is that there are multiple different depressions, and you're not going to find any kind of unifying brain pattern. There'll be different brain patterns for different groups.

- 2.30** This was a study that was done by Kaiser Permanente, where they took mothers who were pregnant going to Kaiser hospitals, and they looked at their records, and they looked for women who were ill during pregnancy. So they had high amounts of cytokines of interleukin-6 or interleukin-1RA. But then what they did is waited. They looked at the kids that were born to these women at ages 9 and 11 and 15 and 17. They found that those with interleukin-6, and they don't specify the type of infection, but they had an infection. They were sick, and the cytokine system was activated.
- 2.31** These kids at 9 and 11 had higher externalizing symptoms, while the other kids with the other interleukin had higher internalizing symptoms. But they both, in adolescence, had higher depression. Depression begins before pre-birth in these kids. So whatever it was that affected that development in utero had an effect on their later life. That's got to be a discrete group. And if you look at them, you're going to see differences in terms of what's going on.
- 2.32** This was a study where they looked at the total population of Denmark again. If you live in Denmark, you've been looked at entirely, ad nauseum. And what they did is they took out anybody who had been diagnosed with cannabis use disorder. Pay attention to this data because we're going to talk about cannabis tomorrow. And what it shows is that individuals who would be diagnosed with cannabis use disorder are much more likely to have unipolar depression, have unipolar psychotic depression, have unipolar depression non-psychotic, have bipolar depression, have bipolar disorder psychotic, and bipolar non-psychotic. And I mean, by a large amount.
- 2.33** I noticed just outside the hotel I'm at, there's a cannabis use store. In fact, there's two of them. Not a good thing. This study shows the relationship of unprocessed food to the development of depression. Particularly, individuals were likely to be depressed relative to how much artificial sweetened beverages they drink or the use of sweet snacks. Now, that's hard to put into any kind of explanatory framework. But it's got to mean something.
- 2.34** We know that depression accompanies all kinds of neuropsychiatric syndromes. It's found in Alzheimer's, Parkinson's, dementia with Lewy bodies, multiple system atrophy, corticobasal syndrome, traumatic encephalopathy, and Huntington's disease. You're likely to find depression. That's not the same thing as the kind of depression we have been talking about. If you have one of these major neurological diseases, and you're not depressed, something's wrong, because your life is ending, and you know it. It's very hard to ignore that fact. So you're going to be depressed. And in terms of treatment, boy, is that different. Because you're actually dealing with how to get a person to understand the reality and to face death. That's a whole branch of psychotherapy.
- 2.35** Finally, we come to social and psychological theories of depression. Interpersonal loss, both psychoanalytically and loss of reinforcer. Learned helplessness and cognitive theories. The psychoanalytic model basically has seven stages, and it's a spiral. It first

was developed by a guy by the name of Karl Abraham, who wrote a book in 1911 called Selected Papers. Sigmund Freud became famous for this theory, which Karl Abraham developed, because he wrote a book in 1917 called Mourning and Melancholia, which is a much catchier title than Selected Papers. I mean, that's the value of a publicist. And poor Karl Abraham just didn't have one.

- 2.36** So it's a spiral. If you know your Freud, you know that there are phases of development. And the first phase is the oral phase, where everything is around the mouth. And what was argued that there was, because of early infantile lack of love, defined broadly, the child stays stuck at that oral stage. So it could be expressed as—I mean, alcoholics were explained by Freudians as being stuck at an oral stage of development.
- 2.37** But in terms of depression, the notion was, you're stuck at this early oral stage and you can't progress into other stages because your needs are not being met. You're very dependent on the person who's providing your needs, but they're not letting you develop. It's usually mom. So dependency evolves. The problem with dependency is that it correlates with lack of self-esteem. In order for the self to develop, you've got to be outside of that oral stage of dependency. So it never, according to theory, it never develops. And that results in the emotionality of love and hate towards what's ever holding you back. But that can generalize.
- 2.38** And then what happens is that that individual is lost. That can be anything, death, leaving, or being there but not being there. What does that emotionality now do? The idea is, it gets turned inward, and that turning inward of that emotionality is depression. So, many hours on a couch and a lot of money, and you'll learn that. But we know that loss is important, but it's more generic. It goes back to, there were studies done around the turn of the century, not this century, the previous century. Let's get it straight.
- 2.39** Kids that were put in orphanages, which was frequent, many of them died of what was called marasmus, which was a general malaise. But these kids were never loved. They're just slabs and treated— a more recent vintage of that is the experience in Romania when Ceausescu was the dictator, and he wanted to increase the population. So he had various stages of status, and your status depended on how many kids you had. And you got paid for having kids, lots of money. So there were lots of kids and lots of unwanted kids, and they weren't cared for. They were stored and fed, but that was it. And they were a very difficult bunch to adopt.
- 2.40** But that kind of separation and this death that occurred by leaving individuals and isolating them was demonstrated in monkeys by Harry Harlow at the University of Wisconsin. Animals go through a response when you isolate them of protest and despair. And the most diabolical is what's called a vertical chamber, where they're basically in a jail cell. It's like a cone, and you can quickly induce severe depression. The

CIA knows that. They know how to deal with people, and they make them very depressed very quickly.

- 2.41 But there may be more of a general thing that most of us are susceptible to. Peter Lewinsohn developed this model, whereby he suggested that depression results because an individual has a low rate of positive reinforcement. And that in itself produces a vicious circle. If you have a low rate of positive reinforcement, you make a lot of verbal statements about being depressed and you have somatic symptoms; you lack social positive reinforcement, and so you tend to be socially avoided.
- 2.42 So where does that come from? Well, it may be that, to begin with, the potentially number of reinforcing events in your life are low. It may be that the availability of reinforcement is reduced. It may be that you simply lack the instrumental behavior to produce positive reward. What's a problem with the kids with a phone? Most of the comparisons that happen, you don't fit. They're neater than you. It's not like—you know how we all tend to find our own levels at some point, right? You find people who are like us. But you don't do that. When you're just looking at influencers, they're ideals kind of thing.
- 2.43 You don't have siblings either. Birth rate's way down. At least the good news of siblings is they knock some crap into you. They knock you around. There's reality. That's the purpose. I mean, a low rate of positive, you go on a date, and if you can't positively reinforce, you go on a date and you talk about applied physics, I mean, good luck. That's just dumb. Applied mechanics now, that's something else. You practice it. Yeah, if you look at what's going on today, the ability to get positive reinforcement that's believable could be in question, and even though this is an old theory.
- 2.44 The other thing about it, though, that depression can also be self-serving because it's hard to deal with a depressed person. You avoid them. And so you're doing the opposite of what they need. They need to be dragged back into a reality. Get out of bed. This forest plot just simply shows how the reward-processing behavior of depressed individuals is very low. It just reinforces that model I showed you of not getting much reward.
- 2.45 Marty Seligman had a theory called learned helplessness, which came from studying dogs that were raised in shuttle boxes. A shuttle box has two compartments, a wall between the compartments, and it's got a grid floor with stimuli bars. So a stimuli comes on that basically is followed by shock that the animal can escape by jumping into the other compartment. And any animal, and you and everybody, would learn that very quickly—get the hell out. Except if you build the wall high so the animal can't get out, and then you take it down, the animal might just stay there. It's like, it's not worth trying.
- 2.46 And that's called learned helplessness. And we see that in some situations of depression, but not all. So people went through a period of looking at norepinephrine

and how it depletes in learned helplessness. And it does, but maybe for other reasons. So it may be only be a subtype of depression. So we went through that with COVID. That was our period of learned helplessness, which we saw dramatically at a political level. People in charge didn't know what to do. They were, as everybody, just making it up as things went along. But what happened is, we had increases in depression during COVID, during that time, but not that severely.

- 2.47** Finally, we come to the theory of depression, which is most universally held. That's Aaron Beck's cognitive theory. That basically is that the depressive has this negative cognition of things, cognitive negative view of themselves, of the world, and of the future. So they view themselves as being inadequate, psychologically, morally, physically. They see the world as insurmountable, putting demands on them that they can't meet. And they see the future as not worth it, full of difficulties. And whatever happens, they're going to fail.
- 2.48** So it in itself becomes a vicious circle, too. You have negative expectancies leading to cognitive distortions, leading to no motivation, no response, negative cognitions, and it just continues. You can see that the major therapy we're going to talk about is cognitive behavior modification. You're going to change how people think about primarily themselves and what could happen to them.
- 2.49** We did one of those shadowing tests with depressed individuals. We also had a control, so we would test anxious, as well as, quote, normals. All of their target stimuli in their dominant ear were just normal, normal stories. The dysphoric distractors, as well as non-dysphoric distractors, were in the non-dominant ear. The distractors, one of them involved themes of helplessness. That turned out to be really distracting to the depressed individuals. It was a direct test of cognitive functioning of what a person will attend to and how it affected performance.
- 2.50** This is just a table, which is a kind of a summary table, which shows how you have to look at to understand the family history of mental health and psychopathology. You have to look at the prenatal environment, maternal neurochemical traits, maternal neurochemical status. You have to look at neurobiological components, early postnatal environment, personality and cognition, and you can see arrows going in every direction. So all of that interacts. So it's not something which is very simple to set aside and understand.
- 2.51** So let's look at treatment. So far, it's been determined that the treatment guidelines for depression done by that—in this case, it was the American psychologist. For children, they say it's insufficient. In other words, if you ask an expert how you should treat a depressed child, they should say, "I don't know," because they don't. For adolescents, then I wouldn't say it. For adolescents, they suggest CBT, cognitive behavior therapy, or interpersonal psychotherapy. For adults, either neuropsychotherapy or second-generation antidepressants. For older adults, CBT or combined pharmacotherapy, group life review therapy. That's "you're going to die. Get over it."

- 2.52** Recently, this model was produced, which shows what's suggested for the American College of Physicians, the American Psychological Association, NIMH, and the Veterans Affairs. They all have criteria for whether you're mild, moderate, moderately severe, or severe. That's determined by that 10-point questionnaire that I told you about. The big winner, if there is a winner here, is cognitive behavior therapy. American College for Physicians thinks that that's what should be used throughout. And so does the American Psychological Association. Interestingly, the psychologists will also accept drugs.
- 2.53** But anyway, this was designed for physicians, so that they would do this 10-point scale, use the scale to determine what intervention should occur. So what happens if you're depressed and you see a GP? That's what's going to happen. You're going to get a script. It'd probably be one of these drugs, but there are others. They fall into a number of categories. First, they're the monoamine oxidase inhibitors. MAO is the major enzyme that breaks down norepinephrine, dopamine, serotonin. It's a major enzyme that breaks down these neurotransmitters.
- 2.54** It's probably the most effective drug. Problem is, it can kill you. So it's not really used. It's used as a last resort. I think I probably took some because it has dry mouth as a side effect. Then there are the tricyclics that actually began as antipsychotics. And again, they're anticholinergic and they have, again, pronounced side effects, but they're not lethal.
- 2.55** Except too much of anything can lead to the suppression of something. That's a primary rule of pharmacology. I don't care what drug you take. You can take too much of it. That includes water. There's such a thing called water intoxication. Did you know that? It's one of the leading causes of death of schizophrenics. Drinking too much on the ward.
- 2.56** Then there's the selective serotonin reuptake inhibitors, or the SSRIs, clearly the winners. Less side effects, although it does affect nausea, insomnia, and there's a sexual dysfunction. Then there are other drugs like Wellbutrin that has side effects, but again, it's the kind of drug you might use if the former don't work. The method of action I've already explained. For bipolar disorder, particularly for the highs, is lithium. You'll see about 78% improvement with lithium. It affects the acute disorder, reduces the relapse rate, and decreases risk for suicide.
- 2.57** Bipolar patients often need to be followed. Periodically, I had a number of patients, and you would be able to ascertain when they were going into a manic cycle. You could then intervene and change medications. That would sometimes be effective. Or they were going into depressive phases. Lithium regulates neurotransmitter signals, a particular second messenger system. So what's the incidence of individuals using antidepressants in the last 30 days? You can see that it's much higher for women than men at all ages. But surprisingly, the age for both men and women where it's used the most is over 60.

- 2.58** So you're dealing with individuals who have lifestyle complaints that are occurring at a point in time where their lives are changing dramatically, and they're being treated with antidepressants. This shows where in the brain you can effect changes. The one thing you know is the top three are slices of the brain in individuals who've had a placebo. The bottom three slices are those who've had Prozac. You can see, the first thing you can see is, the placebo produces brain changes. It's important to know. Placebo is an active drug, even though it's not a drug. The ethical question is, should you just give a placebo? Because it works.
- 2.59** I'll tell you a story. I got interested in the placebo when I was an undergraduate, in the Middle Ages. This was at a religious school. All schools then were religious in one way or another. And alcohol was not allowed on campus. But I was married and living at the end of a football stadium. And so we thought we'd be clever, and we'd sell beer during halftime. Capitalism, right? So we sold tickets. It was really popular. Wads of money. The dean of students learned about it, called us in and said, "If you sell this beer, you're out. You're gone."
- 2.60** Now we'd come to an ethical dilemma. What do you do? There was a brewery about 50 miles away from where this university was, and they produced a product called Nearbeer, which, of course, is universal now. It's no-alcohol beer. But this was the first kind in North America. So we went there and bought cases and cases and cases of Nearbeer. Hauled it back to the university. But you can't sell this. We sold real beer. So we had to go out and get empty kegs, open up all the beer, pour it in, pressurize it up, right? So we did that. Then halftime came, and half the stands just came off. It came to the basement of this apartment building where we were selling stuff.
- 2.61** OK, so we're clear. We're not selling alcohol. What got me interested was not the criminality. What got me interested was the fact that the people who bought tickets, people who were juniors or seniors, said, "What is this shit? Give us our money back." We gave money back to some. But there were freshmen there, some sophomores, who were drunk as skunks, throwing up. They didn't drink any alcohol. That's my introduction to placebo studies. It could have been a study. But it shows you the power of the placebo.
- 2.62** Anyway, you can see here that, in fact, you produce brain effects with a placebo. It's particularly true in depression that there's a large percentage of people who respond to the placebo. The selling of antidepressants has depended on a placebo effect. You take the placebo effect out and you're down to around 10%.
- 2.63** This is another study which just shows the difference in following treatment, and this time in terms of right dorsolateral frontal lobe functioning. This is an important study because it shows you often what happens when your first drug treatment doesn't work. So what do you do? You try another one. And so you can see, if you can look closely, later on anyway, you'll see how they flow from, the dark blue is SSRIs. And all of the rest, most of them are variations on that theme by mixing in other drugs. If the SSRI

doesn't work, you try an SSRI that is mixed with one that affects some other important chemical.

- 2.64** Physical interventions like ECT or transcranial magnetic stimulation are rarely initially used. I'll talk about that briefly. There's this question of, do antidepressants work? I've pretty well answered that. It comes down to about 10% that it works in. When added to the placebo effect, it can be over 25%. Again, large placebo effect. The other thing about what happens with drugs is, if you use the active drug, the individuals, there are a lot of side effects, so people drop out.
- 2.65** Why should you use meds? The psychiatrists will argue because they protect against suicide. But there's no substantial evidence that that's true. Observational studies have discredited, in fact. There are some studies that show that these antidepressants are associated with increased risk of suicide and an increased risk of homicide. So effectiveness is limited. And the major problem with SSRIs is you don't get an immediate effect. You're dealing with a depressed, possibly suicidal patient, and you give them an SSRI, you don't get an effect for at least two weeks.
- 2.66** This graph shows you side effects with drugs, and sometimes they can be very crippling. And there are three bars there because this comes from the FDA, Federal Drug Administration, the equivalent in Great Britain, and from WebMD, where you get the most honest results. And it just shows that when you're taking a substance which is going to rearrange your biochemistry, you're going to have a multiplicity of effects. But again, major problem with the SSRIs, they take a while to be effective.
- 2.67** Consequently, you need something that's going to act right away. It's Spravato, now appears. Basically, what that is, is ketamine. It has an immediate effect. It is being used because it has an immediate effect. That's what these graphs show that you can't see. They show that, within day one, you get a dramatic decrease with the various dosages of ketamine. And that that ketamine significantly affects the biochemistry in the brain. It works fast. The issue is, does it last? And there are side effects: dizziness, nausea, dissociation. But people have said that it works at least for one or two weeks after cessation, in which case the other might kick in.
- 2.68** But there's a whole field of interest in using substances that we know affect relevant neurotransmitters that are currently level one status banned drugs. So in psychiatry now, I've got a number of calls from people who want to start a business where they're pushing some of these drugs, and they know they have to start doing studies. So they're trying to connect all over the place. But you can see how, universally, except for ketamine, how the psychedelics affect the serotonergic system.
- 2.69** Here's a study with a measure of depression using psilocybin, and they get a kind of immediate treatment effect, which was first measured at week 5, but it was dramatic. That's become the go-to study that people are now arguing with governments. Oh, here's another one, where they compared psilocybin to niacin. The problem with using

these kinds of drugs is, it's very hard to do a placebo comparison because the drugs themselves have such acute effects. You know you're taking the drug, and so you're going to give it more credit.

- 2.70** But in this study, it was 11 research sites in the US. 51 subjects got psilocybin, 53 got niacin, and then you can see the difference. There was a drop in depression scores of 19 points for psilocybin and only basically 7 for niacin. These studies are popping up all over the place now. But that leads to questions about, hey, this is still a status one drug, which is illegal. So like marijuana, states are struggling to get approval to write legislation to get these drugs approved, at least medically, for medical use. Different kinds of bills are being proposed.
- 2.71** There's a map of states where currently you can use these drugs. Right now, that's Oregon and New Jersey. And what New Jersey has in common with Oregon, I don't know. But it's going to happen, just as it happened with marijuana. But we don't have to really worry a great deal in terms of asking questions because these drugs have been used for medicine in Switzerland since 1988. So you can look at, and they've been used to treat depressives.
- 2.72** And the stuff I've read gives me the impression that the effect is mostly extra-pharmacological. In fact, people use the word "hyperbole" to express the enthusiasm that's being used. They also argue that the treatment effects depend on context. They're also not likely to be used because they're very expensive. Not that the drug is expensive. You go and get that on the street. But in order to use it medically, it's a trip. So you have to be in the lab for a long period of time. You have to be monitored for a long period of time because there are two possibilities.
- 2.73** One, it's an enjoyable therapeutic experience, or it's just the opposite, and the opposite does occur. So it's not something which you say, "Here, take the drug, go home, and do this." It's done under controlled conditions, which means it's not going to be adopted as quickly as you think. I wonder how true the kind of just statements made by some of these Swiss clinicians happens to be.
- 2.74** The issues are, is it a treatment-specific effect? We know on a cellular level, it increases serotonin transmission, glutamatergic transmission, changes conductivity, increases psychological insight. But there are adverse effects. As I say, it could be a substantial cost. The things we do with depressives, it just boggles the mind.
- 2.75** This was a study done where they just simply used probiotics. So you're depressed, eat your yogurt. It's what it was. They used, I think, 11 different probiotics and got a dramatic decrease in depression. My question would be, who are the depressives you're dealing with? Why are they reacting? It makes no sense. Here's a drug called Zuranolone, which has just been approved by the FDA, and it's for postpartum depression, specifically. And unlike all of the other drugs that I've been talking about that affect neurotransmitter functioning, Zuranolone doesn't.

- 2.76 Finally, I've added this because it shows, if you give kids vitamin D supplementation in the first two years of life, internalistic psychiatric symptoms at age 6 to 8 years are different. Vitamin D. That's all you're doing. And again, it was a study, twins born at term from a hospital in Finland.
- 2.77 There are other ways to deal with depression. And that graph I showed you about these drug effects, it also had ECT on it, but it was a very small percentage in it. It was like, the third time around, we'll do it. ECT is a situation where you apply currents to the brain, typically to the non-dominant hemisphere, where you have two electrodes, you pass current, you produce a convulsion, and you do a series of these. It has remarkable effects on depression. 100,000 Americans are treated with ECT annually. The procedure is far less damaging and invasive than it used to be.
- 2.78 I worked in the back ward of this mental hospital. This was before Jack Nicholson, by the way. So this is really a snake pit. When individuals became too violent, we wrapped them in sheets, tied them down, and it certainly was lacking humanity, but welcome to the 1950s. Patients routinely got ECT, but we didn't have muscle relaxants. So one of the problems with ECT at the time was broken bones because the violence of the convulsion. So they had to take the flunkies out of the ward. I was one of them. And you had to hold the patients down. So, one session of ECT usually required six or seven people.
- 2.79 They got wise at one point, and they started using a muscle relaxant, curare, which gives you the impression you're dying. It just shuts everything down. Now, you don't have to hang onto the person anymore. But it created such a negative experience for the individual. They hated it and didn't want it as a treatment. Today, ECT is done so that you show up in the morning. It's done to the non-dominant hemisphere, so you don't have the side effect of rampant memory loss. I mean, guys that got it when I was there were zombies for days. They didn't know where they were.
- 2.80 So today, it's the kind of thing that if you're suicidal, that's what I would recommend today because it'll clean you up for weeks. It's not a treatment that's going to last. But suicide is, for many, it's momentary. It's not a grinding thing. For some, it's a lifelong experience, but that's rare. So ECT is done today, and you do it and you go home. So it's like minor surgery. And ECT has the quickest action. As I say, anesthesia, muscle relaxants are used. Small currents, convulsions last 25 seconds to three minutes. And you do 6 to 9 treatments over 2 to 4 weeks.
- 2.81 I would like to circle back to the side effects versus the bad trip when it comes to psychedelics. And how is it different from the two? So you mentioned when you take psychedelics, there's potentially a bad trip. But when I hear you speak of the medical side of prescription drugs, it sounds like the side effect is a type of bad trip in its own.
- 2.82 Sure.
- 2.83 For psychedelics, what's a bad trip?

- 2.84** They're very individual. I'll read you a case tomorrow of, talking about drugs, it comes from the 1600s, the Florentine Codex, of a bunch of Indians in Mexico City, getting high on a drug. And the nature of the different responses is dramatic and non-predictable. You get individuals who see themselves doing glorious things. But moreover, you saw individuals who were being sacrificed. They were being killed. Now, how would you predict who's going to have one response versus the other? We don't know.
- 2.85** But I think you're right in terms of drawing the analogy, because we don't know who's going to respond negatively to pharmaceuticals. So the analogy holds. I mean, it would be great if we could find that out before you wrote a prescription. I would know enough that, if I asked you if you taken drugs before and had negative experiences, and you said yes, I would say, aha, you're more likely. And then I guess it would be, the decision would be up to the doctor and you to decide what to do. But I don't think that's even considered. You see a GP today, how much time do you get? Don't they have revolving doors now? Or trap doors? You're gone. Yeah.
- 2.86** You mentioned depressives having lower self-esteem, more negative outlook on life. Would religious systems that elevate self-worth or maybe give a more positive outlook on life make you more resilient to depression?
- 2.87** That's not a question. The answer is obvious, right?
- 2.88** Is there any statistical findings in that?
- 2.89** Well, yeah. That's why before I mentioned in that happiness thing of people who are married are happier. People who have a normal life are happier. People who have belief systems are happier. If you're a liberal, you're not as happy. Why? Because you're all over the damn place. You haven't got anything to nail your feet down with. No, you're just giving me a—this is my life. This is what I believe. This is the way I want to live. So you have a roadmap. And there are roads. If you don't have that, you're on a dirt road in the country where you don't know where you're going. And that's stressful. Absolutely.
- 2.90** From what I understand is that you have worked with Dr. Peterson on creating the Future Authoring program and the personality assessment. I guess, what I want to understand is, it seems like depression and other forms of things that negatively affect you in your life mentally stem from lack of—and jumping off of what he was talking about, it's like a narrative framework, and that's one aspect of it, not the only. I'm curious, from what you've learned and studied, how much do you understand how that narrative, where does that narrative framework come from? How do we—where are the origins of that? How is it that it's—
- 2.91** You're stuck by where you're born. That narrative is handed down by many. We go through periods where we revolt against it. And some of us wise up and realize, hey, there's some wisdom here. Some of us don't, but we continue to fight at windmills.

Lecture 3: Recognizing Risk

At a glance — Recognizing risk.

- 3.1 This shows the effectiveness of ECT at various dosages on a scale measuring depression. It shows that you get the most effect with the highest dosage. But that's life, right? You want to—well, anyway, this figure just shows you dramatically how ECT, which is done to the non-dominant hemisphere, looks in comparison to using magnetic seizure therapy. So far, all the stuff that I've seen relative to these two is that ECT is clinically far more effective, but it's not as palatable to patients, or as magnetic stimulation is.
- 3.2 Now, magnetic stimulation doesn't cause seizures, which seem to be required for an ECT effect. This is just a figure which shows the effectiveness of bilateral repetitive TMS in older adults, and it's not that dramatic, which is why you would see a more dramatic effect with ECT than TMS. But TMS, again, is used more in clinical practice because it's in and out. And basically, anybody can do it. You don't have to be an MD to do it.
- 3.3 If you look at ECT in comparison to ketamine— this is a forest plot that shows you that in fact, ECT is far more effective than ketamine in reducing depression. But it doesn't last. We don't know if ketamine treatment will last or not—just hasn't been done long enough.
- 3.4 Finally, and quite recently, it's been shown that there are at least two types of depressed patients. So this looked at over a thousand patients, gave them a battery of psychological tests. And these tests measured all kinds of cognitive functions, but heavy on the frontal lobes. You can see how one group of depressives is really deficient compared to the other. That group makes up about 28% of depressives. You look at something like this and you say, "Well, don't do cognitive behavior therapy with them, because they're not functioning that well cognitively, period."
- 3.5 Furthermore, when they were looked at in terms of many imaging studies, it was really the frontal lobes, the prefrontal lobes, and the cingulate cortex—which is just back from the frontal lobes and overlaps around it—were producing the effect. That's another point of: there's another group of depressives that deserve individual attention. But don't do cognitive therapy with them. But cognitive therapy has benefits. It doesn't interfere with medical treatments. There are no physical side effects. It's often preferred by patients, rather than drugs. It provides patients with tools to help cope with future symptoms, which may reduce relapse.
- 3.6 The premise of cognitive therapy is that a person who is depressed has a maladaptive way of looking at the world. That's Beck's theory, on which this is all based. And so hence, they process information in an unhealthy way. So what you want to do is correct the way the person thinks. And you point out the nature of thinking errors and lead them to put things into a more appropriate perspective. This effect is enduring because

if you can change how a person thinks about things, you're changing how they interact with the environment and things around.

- 3.7** And relapse rates are lower following acute CBT than drugs. And it's lower for a combined treatment versus patients exclusively treated with drugs. They have reduced relapse, 72% to 46%. You can even boost that by doing what's called mindfulness behavior cognitive therapy, where you learn to teach how a person's thinking about things, to shift their thinking in a positive direction, or to shift their thinking actually towards what's making them depressed, but teaching them how to handle it. This figure just simply shows you the effectiveness of CBT preventing relapse versus drugs. And this is another study where if you add the mindfulness component, you also can get dramatic effects. So there are procedures to deal with depression that do not include the physiological.
- 3.8** What's happening now is the internet's filled with apps to moderate depressive symptoms. This forest plot just simply shows the relative effectiveness of internet procedures that specifically are designed to do that. You can see that the results vary, and they vary dramatically. You can get a number of programs that will actually make you worse. There's no way of knowing which program is going to be more effective than others and whether the single finding is spurious.
- 3.9** One of the most recent, which was approved by the FDA, is called Rejoin. It involves you doing a bunch of manipulations. It's designed to stimulate the amygdala and the prefrontal cortex. This has been approved by the FDA, but the data shows that it doesn't work. FDA approved it because they feel like it doesn't hurt anybody—they actually said so— but it doesn't work. In which case, I'd argue it hurts somebody. This is data from a computer program called Woebot that will treat your depression. They put out scattered data, but you can't see—they stopped doing it because—you can look them up on the web.
- 3.10** Finally, two things. This is a summation which shows that you can decrease depression by walking and jogging, by yoga, by aerobic exercises, by strength training, by tai chi, and by various combinations. I've always found that the one thing you want to do with a depressed patient is get them active. I've also found that the hardest thing to do is to get them active. That's the essence of depression. I don't know how they got these people to walk and jog.
- 3.11** I even tried playing tennis with depressed patients. "I'll meet you at the court, and we'll do this, and we'll do that." So who's the fool? It didn't work. But you know, there's a person who's sitting at home doing nothing. You know you want to activate them. You want to get them out doing things. And so you think, logically, this is what we have to do. Doesn't always work. So people have been using combinations of pharmacotherapy and psychotherapy. This is a forest plot that, again, shows minimal effect.

- 3.12** Finally, apropos to what we've been discussing: I want to deal with depression or actually prevent it. Try eating meals at home with a family. Try being a family. Try providing some kind of glue which is your own society and your own norms. Do it in a way in which it's not distracted. So turn off the televisions that are running right around the kitchen or the dining room, and try talking to each other. That's a behavior which we seem to have lost. Many of us have lost. It's the essence of what social animals should be doing, rather than being lone individuals.
- 3.13** But of course, there are many types of depression, which is the point I've been trying to make. In this working as a family, I think we'll deal a lot with the current spike that we're seeing in young individuals. But you're going to have to start younger to shape it up because you're going to have no luck dragging a teen to a dinner table today and expect to have a reasonable conversation. I don't think that'll work.
- 3.14** So what to conclude. There are multiple depressions. They probably each have different pathways that lead to that behavior. From the case that I started with, she had no inkling of what was happening, but it was clear that there's a family relationship, and likely that the stress of university at that particular time kind of interacted. So that's the causation. It probably points to some kind of physical problem to begin with. But that's only one time.
- 3.15** Causation itself is likely multifactorial. We do know some general risks, and I presented them, but much more remains to be elucidated. And treatments will markedly improve once differential ideologies have been determined. Then we have this quote from George Bernard Shaw. "The secret to being miserable is to have the leisure to bother about whether you're happy or not." That's true, too.
- 3.16** You know what bridge this is? Right. It's the Golden Gate Bridge. And what's unique about it? [inaudible student] Okay. It's the symbol for jumpers who are committing suicide. It averages 30 a year, of individuals who have committed suicide off this bridge. Hence, what was done is you can see that there's a net around it. It's a steel mesh net that's just been built to try to prevent these suicides. The price of the net was supposed to be, I think \$140 million. It's up to \$400 million now, and growing. So there's a great debate about whether it's an appropriate source, of an attempt to prevent suicide.
- 3.17** I present it to show you that although suicides are small in number, actually, societally, they have a tremendous impact, and they affect how we look at a lot of things. Because there was a great deal of argument that one should look or spend that money, for example, in mental health in general. That debate lost, and the money has been spent, or is being spent, or is in court, one of the two. It illustrates the significance of the topic per se.
- 3.18** Yeah, an average 30 individuals each year jump from the Bridge. The survival rate, by the way, is less than 5% because it's a long fall— was, now you hit steel. It was a long fall, and there's sharp currents in that San Francisco Bay Area. And contrast that cost,

and the fact that only 2% of suicides involve jumping, whereas 50% involve firearms. And yet we don't put steel nets around firearms.

- 3.19** So originally, I phrased this talk as "Suicide et al. Why? Why not?" I thought the why not was rather provocative until MAID came along. Do you all know what MAID is? [medically assisted suicide] Yes—medically assisted suicide. In fact, Canada now leads the world in medically assisted suicides. The number of deaths in Canada, from June 2016 to the end of 2022: 44,958 individuals were killed. People don't like, a lot of people don't like me to put it that way— but they were killed. At that rate, it will be 65,000, in 60 years, per year. That's more dead—currently, 44,000 is more dead than all the soldiers Canada lost in World War II. So it's not trivial. It was 4.4% of all deaths in our last report. In Quebec, it's 7% of deaths.
- 3.20** Two issues that arose. One, sarcastically, is: that's how Canada is dealing with a health crisis. But that's sarcastic. The other is its relationship to the collection of organs. And that's a real issue, because you're starting to see stuff of "when is a person dead and when aren't they dead," et cetera. At the current rate, it would be 65,000 per year.
- 3.21** Mental illness was recently, specifically, excluded until March 17th, 2023. If you were on that liberal committee that was doing this, why would you have excluded mental illness? If you've been listening to what I've been saying, it's because we don't know what it is. It's pretty hard to make a judgment about whether a person should die or not. But that's a treatment that works, it does. But at what cost? Because our subject is dead, we learn a lot from statistics. So I'm going to give you an overflow of graphs.
- 3.22** This is the Canadian death rate by suicide. The first thing, the blue is male and the red is female. So the first thing you see is the death rate is expressed per 100,000 population. All the statistics will be the same: per 100,000. You can see that there's a peak in Canadian death rates for males between 50 and 64. That's not the same as it is in the US. In the US, there are far more male suicides than female suicides, but it's over the age of 75. I read the last report for MAID, and the vast majority of the age was over 70.
- 3.23** This is just a comparison to that because, again, the red is female, the blue is male, but it's not suicide. It's hospitalization for self-inflicted injury. If you look at self-inflicted injury, it's young women ages 10 to 19 that are so predominant. You see that in a graph that I showed you about depression, which is basically this same figure. It's funny the fact that more males commit suicide than females. We see this dramatic uptick in women, girls, 10 to 19, 20 and 30 years old.
- 3.24** We can look at suicide by race and sex. And what you find is that it's high in AL, which are American, Indian, and Alaskan natives. And it's high in white race. Here's another older figure of Canadian data for Aboriginal males and females. Again, for the age group of 15 to 24, it's Aboriginal males that show a dramatic, astronomical, increase in suicide. This figure shows the degree of stress faced by a white population, and by

American Indians. You can see that there's a dramatic difference in the level of stress, in its all-time daily stress, that the American Indians and Alaska natives affect.

- 3.25** So who commit suicide? Elderly males in Aboriginal groups and Native Americans. They also have exceedingly high levels of liver disease and other diseases. Because—this is a forest plot that looks at the relationship between alcohol use disorders and suicide mortality— and what it shows is that there's a high correlation between the two.
- 3.26** So more data of, in this case, suicide attempters. 12.2 million adults—this is US data— each year have suicide thoughts. Of those, 3.5 million will make suicide plans, and 1.7 million will actually attempt suicide. And of those attempters, there are some that make the plans and some that do it just impulsively. Although suicide itself is something somebody may be thinking about a lot, but it still is an impulsive act. Whether it works or not is going to depend on the method. I haven't labeled male/female here, but it should be obvious. It's the males who use guns. 57, 58%, whereas females are much more likely to use poisoning, than are males.
- 3.27** Then we have data which shows that it varies by profession. If you do it by work status, people who are oil roughnecks, and miners, have actually the highest rate. But if we look at professional standing: medical doctors, dentists, police officers, veterinarians. So what's in common? Why should a medical doctor have a higher suicide rate than a real estate agent or a financial services person? It's dramatically different. Isn't that a very high status job? Anybody know?
- 3.28** Maybe because they witness more death on the job?
- 3.29** More stress?
- 3.30** Yeah, well, certainly more stress.
- 3.31** Lots more stress because you have lives in your hand. The dentist, you have mouths in your hands—and they bite. Police officers, well, any status you had, you've lost. You can kind of make sense out of what's going on.
- 3.32** But for example, the medical profession itself is changing. You have all this responsibility, but you have very little control. Somebody tells you how much time you can spend with each patient. You spend two-thirds of your time doing paperwork, and yet you're responsible. Somebody's telling you how much time you can spend. You're limited in terms of what you can do and what you can learn. The idea of being a family doctor, for example, is filtering away. Dentists have always had a high suicide rate. If you don't want to go someplace or something's wrong, what do you say? "It's like going to the dentist." There's nothing more aversive. So they may take it to heart.
- 3.33** This is an unusual graph that I found because it's suicide rate and suicide thoughts in autistic individuals. It's totally different for girls and boys. Autistic girls are much more prone to suicide and suicidal thoughts than are autistic boys. When you compare them to autistic kids who are not suicidal, which is the bulk, and you ask the mothers what's

going on different in their lives, it's often the kids who have been bullied. But why is it so different than what you find in non-autistic kids, where the curves flip? It's hard to explain. But again, it may go back to more susceptibility of females to suicide. The study didn't break it down in terms of just successful deaths versus just self-harm events.

- 3.34** You can see this is an older graph, but it still exists and still relevant. It shows that suicide rates have gone up, but so have accidents. Which begs the question: are a lot of accidents that occur suicides? I did a study with Transport Canada years ago, where we looked at single-person car crashes and tried to do psychological autopsies. We did kind of half-assed autopsies, by just looking at records. What we found was inevitably, the person drove into an abutment on a major highway, was intoxicated, and had a substantial problem that day or week before. So it's been labeled an accident. But increasingly, we looked at it and said, "Uh-uh." This guy just wiped away his problem, whatever it was, because it was discrete that you could find an issue that had occurred in that individual's life.
- 3.35** It varies geographically. Can you see which country has the highest suicide rate in the world? 72.4 per 100,000. [inaudible] Pardon me?
- 3.36** Lesotho?
- 3.37** Lesotho. That's not a car. That's a country. Where is Lesotho? Anybody know?
- 3.38** Southern Africa.
- 3.39** Yeah, it's a small, surrounded by mountains, encapsulated by South Africa. It's in South Africa. It used to be a chiefdom. I think Swaziland is there, too. It's down of some place. Anyway. Yeah, Eswatini is Swaziland. It's the only kingdom left on Earth—pure kingdom. Anyway, so they have the highest suicide rate, and the issue is why. And so they've been asked. And they say it's a lack of mental health help.
- 3.40** But it's not really what's going on. I mean, they can't deal with it because they don't have enough people—it's a population of 2 million, and they have, what was it? Fifty doctors? That may be stretching it. One-third of them work in South Africa. Because that's where the money is. But they have a terrible problem with AIDS. And so that's where all the medical attention is.
- 3.41** Now, why would they have a high suicide rate? Well, possibly because there's a lot of sickness. A lot of people are left alone. But it's also a country in serious transition, where many of the men used to leave the country to work in the mines of South Africa. And those mines are petering out. So they no longer work in the mines, and they've come home but have no status. And that turns out to be a rather significant variable.
- 3.42** The second country is Guyana. What do you know about Guyana? Nothing. Who is Jim Jones? 591 people committed suicide. Now, that doesn't appear in Guyana's statistics. Guyana's statistics are the way they are, and they're strange because in this graph, they're second with 40 per 100,000. It's mostly Indo-Guyanese farmers who kill

themselves by using overdoses of pesticides. But this year, they have zero. Next year, they may have 40 per 100,000 again, but they've decided not to play the statistics game. Anyway, it varies by country and that can be explained somewhat.

- 3.43 Here is something which I think is really relevant to us: a survey of 190,000 US adults on suicide ideation. They looked at where you lived and your sexual orientation and your race. What they show is in each of those figures, you can see an increase. It doesn't matter where they lived. It doesn't matter if they lived in a Metropolitan area, a small Metropolitan area, or a non-Metropolitan area. All the figures look the same.
- 3.44 What increasing is not race, but one's sexuality. If one is either gay or lesbian, you get an increase. If one is bisexual, you're at a high level. It's the same statistic no matter where they live in each one of those graphs. So sexual identification—and the way that's been taken, of course, is to say that it's the universal prejudice which has, in fact, isolated these individuals. That's certainly partially true. I don't know if it explains everything. But will acceptance change? That remains to be seen.
- 3.45 It turns out that people who are cyberbullied are more likely to have suicidal thoughts. But that's true of people who do the cyberbullying. So it's not just that you're, "the victim." If you're also the perpetrator, you also have such thoughts. So there are reasons why people are cyberbullies, and it's their own problems that they're trying to foist onto others.
- 3.46 We've seen these depressive symptoms go up. Both these figures show that. The first is 9th and 12th graders, and depressive symptoms increased dramatically for females, and hours spent on electronic devices has been increasing dramatically. You can't read this, right? It's a bunch of little slides that show trajectories for psychopathological and behavior symptoms.
- 3.47 Remember I said trajectories are important? Well, this looked at the trajectories for withdrawn behavior, somatic complaints, anxious, depressed, social problems, thought problems, attention problems, delinquent behavior, aggressive behavior. Which of those two do you think actually predicts suicide? Turns out it's individuals who are withdrawn, and it turns out it's individuals with somatic complaints. So we're talking about people who are alone, psychologically, and can't test their ideas and thoughts.
- 3.48 Suicidal behavior is also most seen in the context of mental disorders. We have: mood disorders, the bipolar major depression, psychotic disorders, anxiety disorders, substance use disorders, eating disorders. It's found in all of them, but not to the level that you can make accurate predictions. You can see that here. The dark blue is the percentage of individuals in each of those disorders, which is related to suicide. It's almost nonexistent. Just like 2% for anorexia, 5% for bipolar, 5% for schizophrenia, 3.5% for substance abuse. That's in women. For men, it's a little bit higher in disorders.
- 3.49 This group, the US Preventive Services Task Force, was asked whether it's worthwhile to screen for suicide. They concluded: no, it's not. There's not enough evidence that we

even have a screening instrument. The benefits of early detection would be helpful, but because we don't have the instruments, you can't do it. So they say at the present time, it's not worthwhile.

- 3.50** Nevertheless, the DSM has proposed suicide behavioral disorder to be included in their next edition. And to be included, you'd have to have a suicide attempt within the last 24 months, it involved not in suicidal self-injury, and it did not be initiated during delirium. And finally, the act not undertaken solely for political or religious objectives.
- 3.51** But this study, which you kind of can't see, was machine learning to see if they could predict suicide. It was marginally successful, but the stuff that it picked out makes no sense, which I think we're going to learn a lot from AI that way. For example, the most predictive thing for women was their socioeconomic status. The second most thing was their father's age, the mother's highest level of education, level of positive interactions, their gestational age. Again, general—you add enough variables together, you can predict anything, but it's meaningless. And that's what this study proved.
- 3.52** Here's one that looked at just some measures of suicidality, the scale for suicidal ideation and family history, both of which were correlated significantly with suicide.
- 3.53** You can look at maps of the US, and suicidality varies with geography. It's more prevalent in the West. Also on this map is suicide that's B, drug poisoning, A, and, C alcohol poisoning. You can die a number of ways, but suicidality and drug poisoning are highly correlated, as is alcohol. I was wrong about drug poisoning— that is in the old Rust Belt. This figure just simply shows how suicidality is associated and correlates with drug poisoning, suicide, and alcohol deaths in men and women at different ages and at different races.
- 3.54** Let's look a little bit at the history of suicide because it in itself is enlightening. This is from a letter written by a poet to his mistress in 1860 in London. "A man was hanged who had cut his throat, but who had been brought back to life. They hanged him for suicide. The doctor had warned them that it was impossible to hang him as the throat would burst open and he would breathe through the aperture. They did not listen to his advice, and they hanged their man. The wound in the neck immediately opened, and the man came back to life again, although he was hanged. It took time to invoke the alderman and to assemble, to bound up his throat, below the neck, until he died." That is a situation where suicide had a weird barbarity.
- 3.55** It was not uncommon in that time of enlightenment. In England, it was not unusual that a suicide would be taken to the town square and hung on a gibbet. A gibbet is just a meat hook for display, as you would a criminal. Or it was also prescribed to be taken to the crossroad and to be a stone put over the face and a steak driven through the heart, which is quite a contrast with today and how we view suicide. We see suicide in a different way. So the perspective has been changing and is changing.

- 3.56** So now let's look at some fallacies about suicide. And by the way, that history is in a book by a guy by the name of A. Alvarez, who wrote a book called *The Savage God*, and it's beautifully written. Because he's a poetry critic for the London Times—was. And he was so taken with the death of Sylvia Plath—the poet—that he really studied the subject.
- 3.57** Here's some fallacies. First is that suicide is inextricably mixed with young love—Romeo and Juliet. No, wrong, fake. The youth are great attempters. They're seldom successful.
- 3.58** An act of suicide is an act of great passion. You'll see it's often the statistics that it's basically the elderly that are more likely to commit suicide. The London police believe that they can ascertain who has died for passion and who hasn't, because they look at the hands of the corpse, and if it's all cut up, then they say the person died for passion because they jumped, changed their mind, hit the water, swam to the abutments that hold the bridge, and cut themselves because of all the barnacles that are on the bridge. They tried to save themselves. The one that dies for financial reasons, they were straight down.
- 3.59** Caused by bad weather. You would think that we are deeply in trouble here in Canada if it's caused by bad weather. But it's not. In fact, there's no correlation with winter. The highest rate is in spring. T.S. Eliot has a poem about April and how we respond to the change in spring. It's the contrast effect which increases the likelihood of suicide. In a time when the bulk of the population is feeling better, the suicide's turmoil just explodes. You'll find a little peak around Christmas when most people have more directionality and it tends to be a happier period. It's during those periods that you see this contrast effect.
- 3.60** Finally, that it's a national habit. We often would blame countries that had high suicide rates. But the one thing you learned quickly is, until relatively recently, it was really fake, the statistics. For example, Italy had a very low suicide rate for a long time, and so did Ireland. Why? They're Catholic countries. I read a report from one coroner in Ireland who was at his wits' end for lying about how people die because this person had put a gun in his mouth and shot himself. And the coroner wrote, "And sure, he was only cleaning the muzzle of his gun with his tongue." I mean, making fun of the fact of what he had to say because the church demanded, I mean, if you wanted to be buried and not excommunicated, you would not commit suicide. Or the church could not know.
- 3.61** Then there's this issue of "those who threaten, never do." Well, approximately 75% of suicides will give you some warning, maybe even higher, because there are different types of warnings. I'll get to those later. And 10 to 13% of attempters will be completers. Furthermore, the analogy that's used is like diving off a high dive. Once you've done it, it's not such a big deal. You can do it again— unless you landed on your

stomach, then you won't do it again. But it's been said by people who've done it, attempted multiple times, is that it's easier the more attempts you make.

3.62 So more fallacies: that suicide notes explain all. I mean, what's more communicative than a body hanging from someplace, than a note pinned to them? But only one-third of individuals actually write notes. A friend—an ex-friend of mine, he's dead—Ed Shneidman, who started the first suicide clinic, intervention clinic, with Norm Farberow in Los Angeles. He got all the notes from the LA coroner, and he analyzed all the suicide notes. Basically, what he concluded is that notes frequently don't explain reasons.

3.63 He said notes fall into five categories. There are notes that express a particular viewpoint, more so a thesis. Percy Bridgman—he was a physicist—he wrote a note, which is really a treatise on why suicide should be accepted. Then there are people who write notes that are against something, that they're upset about. That's the antithesis. Then there are notes that are antithetical—they have no point of view. Like a note which is just a list of all the possessions that a person has and says, "Now, this goes to so and so." Then there are notes where people put together new insights. That's a little like a thesis. Finally, there are conflictive, empathetic notes. The most dramatic was one which said, "Dear Mom, I hate you, love John." And hanging on his body. I mean, you can't get more communicative than that in terms of the message that, in fact, you're making.

Lecture 4: Reducing Risk

At a glance — Reducing risk.

- 4.1 We were in the middle of talking about the difference between attempters and completers of suicide. It's only about 10 to 13% of attempters, in fact, will be completers, which is why it's so hard to predict. But it's important to look at the fact that the gesture itself, as a form of social communication, may not necessarily be linked to suicide, but is, as Professor Stengel said, a cry for help in one way or another. I'll give you two examples to show you the extremes.
- 4.2 The first was a case of a young girl, about 17, who was having great acrimony with her parents because she was dating a guy who was in his mid-20s, and that's a catastrophe in a lot of homes. And so she would continually fight with her parents and one day decided to take a bus from the suburbs where she lived to the city, stopped at a drug store, bought some razor blades, walked up the mountain, Montreal, to a hospital, and went into the ER and just nicked her wrists a little bit. And so they bled a little bit.
- 4.3 And she had this vision of what would happen is that the physicians and the nurses would run to her assistance, call her parents, who would rush to her side and forgive all of her problems and allow her to be with her boyfriend. Sadly, that's not the way medicine works in Quebec or in Canada. She sat for about a half hour or so and got totally disgusted, basically said anybody could die here, and of course they have. She went back to that drugstore, bought bandages, and got on a bus and went back to the suburbs. But that's a suicidal gesture, but without lethality. But nonetheless, it's a message of some difficulty.
- 4.4 But here's another case. This is a woman in the 'burbs who was in her late 30s and had the feeling that she was becoming very estranged from her husband. Her friends had moved away, and she really had nothing of a life in her suburban home. Her husband worked in the city and took a train there every day, and he was very busy, but he was quite compulsive. He always arrived home, would take the 5:30 train. He would get home by 6:00. He'd be walking in the door.
- 4.5 She wants to give him a message. What she did was, she went into the garage, put towels around the base of the garage door, started the car, sat in the car, let the carbon dioxide, monoxide, grow in the car, in the garage, knowing that her husband would be home at an exact time because it was very compulsive. Sadly, he missed the train that day, and she died. And so the issue is, where do you classify that kind of suicide? She didn't really want to die. She was giving an extreme message to her husband. And it's sadly, because of the circumstance, she died.
- 4.6 So the gesture can be a form of social communication; can be an operant. And you can see this when we compare attempters to completers. Women are much more likely to be attempters than completers. They are apt to be younger while completers are older.

They use low lethality versus high lethality. Pills, wrist lacerations, versus firearms. There's a big difference in terms of whether one's going to be successful or not. The setting is usually a chance of rescue versus a low chance of rescue.

- 4.7 The nature of the disorders, in fact, differ. Borderline personality disorder is a frequent concomitant of borderline personality disorder is our suicide attempts. Whereas you're apt to see more discrete diagnoses of mood disorders, schizophrenia, substance abuse, in individuals who complete suicide. We want to try to at least prevent suicide in some individuals. Now, remember, in terms of what we've been discussing, there are many different types, with a large number of individuals who commit suicide, who are elderly, sick, and who basically fall into the category of where MAID is now operating, mostly.
- 4.8 But there's this phrase, which I think is really relevant when you're talking about suicide for many individuals, and that is, it is a permanent solution to a temporary problem. People who attempt suicide need to have that idea cemented in their brain because most feelings of disaster pass. But there are even legal aspects. It used to be that suicide, in fact, was a crime. It's no longer a crime in the US, but it is the second reason most involved in why psychiatrists and physicians get sued, particularly psychiatrists. That is, they will be sued for not predicting and preventing suicides.
- 4.9 Because we'll see, I'll show you later on, that the individuals who have been hospitalized for a suicide attempt are extremely likely to actually go through with the act the first few days after they've been discharged after treatment. It's the psychiatrist who then gets blamed for that temporality. Then, of course, there's the effect on the survivors. There was a survey done for survivors, and you get a variation in response. In this survey, 8% of the survivors felt relief.
- 4.10 I know of such a case. I knew the kids of this woman. They were all in trouble. They were all truants and all being treated in one way or another for basically a lot of antisociality. I ran into them a few years later, and they were all doing great. The issue is what happened in the interim. The mother walked in front of a cement truck, and it made a startling difference in her offspring. But that's a very small percentage of individuals. For the vast majority, the individuals, can't read my writing, suffer in one way or another.
- 4.11 In fact, 67% of individuals close to the individual who commits suicide, become depressed. That's the— and another 25% develop health problems. It has a pronounced effect on survivors. What are some of the common predictors of suicide, mental illness, depression, alcoholism, drug abuse? Yes, yes. Suicidal thoughts, talk, preparation, prior suicide attempts, of course, the lethality of the methods, and then individuals living alone without social support. We'll see that social support is probably the best preventative for individuals who are thinking of suicide.

- 4.12 You can use certain kinds of measures that have a little bit of predictability. One of these is the Beck Hopelessness Scale. If you read the kinds of things that it asks, you have to answer true or false. There's no use in really trying to get something I want because I probably won't get it. Things just won't work out the way I want them to. When things are going badly, I am helped by knowing they can't stay that way forever. If you have no optimism, then you are at risk to begin with, and these tests can sometimes elucidate those things.
- 4.13 Common single predictors, older white men, modeling it, that is, it runs in families, genetics, work problems, any particular issue. Often, it's individuals who have been pushed up against the wall because loans have become due and they can't meet them, etc. Marital problems, stress, life events, anger, aggression, low serotonin, and physical illness. This is the data I wanted to show you of individuals released from a hospital. It was done in Finland, and they followed up almost 10,000 people. No, more. 91,000 of treated depressives, and they looked at who would, in fact, commit suicide and when.
- 4.14 You can see from the curve that it occurs very closely to when they've been discharged, and it drops dramatically. That colored slide on the right is a heat map, and it just shows you where all of these suicides occur. The days and the first few week, basic first week of discharge, from being treated for depression or a suicide attempt. It's like they've been using the hospital and treatment for their planning.
- 4.15 These various slides basically track a sample of those individuals to show the few in which it actually increases. Those few are individuals who have alcohol use disorder or a substance use disorder. They're the individuals who don't conform to that rather immediate response. This is a study which classifies suicides. It looked at US suicides, over 300,000. It came up with these five classes. They were able to categorize all suicides into these five classes. They are mental health and substance problems, just mental health problems, alcohol-related and intimate partner problems, physical health problems, and poly-substance problems.
- 4.16 Now, which of these do you think is the most severe and accounts for most individuals? I can show you here. It's Class IV, which is physical health problems. All you have to do is look at the statistics they've made now in Canada. And what is the group that's most likely to commit suicide? It's the elderly. It's the sick and elderly. And that is followed by Class V, which is poly-substance problems. You can see how physical health problems have been consistently high and predictive since 2003 to 2020.
- 4.17 But what is increasing are these poly-substance problems, and that started around 2012. Decreasing are the two mental health problems. They are less and less predictive of suicide, as is relationship-type things. So even suicide can be categorized. The problem with all of this— I'm going the wrong way. The problem with this is, most high-risk patients don't attempt suicide. Correlations are with groups, not with individuals, and there's a very low base rate. As Cesare Pavese said, "Everyone has a good reason for suicide." But very few act.

- 4.18 And for some, it can be what's been called a vocation, a closed world. But we talked about that the other day, where, for some individuals, it's just always there as part of their life. Related to suicide is something called murdercide. These are people who will kill strangers for ideological reasons. Studies have been done, for example, of 400 Al-Qaeda members and, quite opposite to what we've been saying of a person who's alone, these individuals can often come from upper or middle class, from caring families, be college-educated, and be married. But they get so involved in an idea that it supercedes normal control kinds of mechanisms.
- 4.19 To understand that, we can look at some really classical theories about suicide. That's the work of Emil Durkheim, who was a sociologist around the turn of the 19th century. He said there are three classes of individuals who are involved in suicidal behavior. The egoistic, where a person doesn't belong. They don't fit any group. They're on their own. They don't have any glue to their relationships.
- 4.20 Then there's the altruistic, where there's a minimal distinction between individuals and the group, which is like these Al-Qaeda people I just talked about, or I think I mentioned the Jonestown Kool-Aid venture, where 500 and some individuals committed suicide in Guyana in the 1970s. And there have been others, collective groups, where the groupthink becomes more important.
- 4.21 There's, what was it, the Hale-Bopp comet that occurred a number of years ago, where people thought that it was a particular comet that was coming. So they knew they would have to die to catch the comet. Bear with me. So they all stacked up in one room. And there was, I think, close to 20 people that committed suicide so that they could be dead to catch this comet and ride forever into whatever. That's when the group becomes so powerful that it absorbs your individuality.
- 4.22 Then there's an anomic type where change has occurred which is so dramatic and so quick that people just simply can't cope. There have been a number of people who have, quote, struck it rich, won a lottery, and then committed suicide because they couldn't cope with the change in their lifestyle that had occurred so rapidly. Often happens with individuals who obtain fame very quickly.
- 4.23 We have a number of other theories of suicide. There's the analytic theory, which is "thanatos", which basically says—you see, Freud could not cope with the horrors of the First World War. He had this concept of eros, or the living instinct, on which the whole theory developed its energy because it's a closed energy system. But the wanton destruction that was occurring in the First World War, and in all wars, he couldn't understand. And so he had to create a drive leading to destruction. And that drive he called thanatos.
- 4.24 And if it remains unconscious, it's going to be people who are suicidal take risks, self-mutilate, have poly-surgery, major accidents. And if it's conscious, it in fact can produce suicide. The problem is, like most Freudian thinking, it's circular. That is, a person does

risk-taking. Why? Because of an instinct. How do you know that's an instinct? Because they did do risk-taking. It just simply is not informative.

- 4.25 Or you can see it as a magical act involving rebirth. You often can see this in some schizophrenics. A young, a young schizophrenic boy would cut himself and bleed rather profusely. And he would say, "I'll live when you're all bled out. So my parents and sister bleed out most when I cut myself." So it's you that, in fact, are being affected. And it's very close to what people have called world destruction fantasies.
- 4.26 For example, Housman has a poem. It says, "Do you creatures, do you love your lives? And have you ears for sense? Here is a knife, like other knives, that cost me 18 pence. I need but stick it in my heart, and down will come the sky, and Earth's foundations will depart, and all you folks will die." So I'm not killing me, I'm killing you. And you're going to go. Schizophrenics can have that a thought process.
- 4.27 Then there's despair, real loss. If you read Huckleberry Finn, Tom Sawyer, his relationship with his stepmother or aunt, it is very much of that theme. Finally, to expiate guilt is another reason for. You finally end up with a theory such as this, where suicide gets explained by various risk factors. I mean, all psychological theories integrate the genetic family history, biological factors, and personality traits. And so those all interact, and they all interact to affect the probability of engaging in some behavior.
- 4.28 If there's sufficient, then you can go over a threshold. If there's other precipitating factors present, then that can lead to suicidal behavior. But you see, there's also, in the middle, factors which should be inhibitory, such as being cognitively flexible, having strong social support, lack of precipitating life events, no losses, hopefulness, and perhaps even treatment for a particular problem. We have the typical biological factors that tell us about this self-destructive behavior.
- 4.29 There's a number of disorders that are genetic that typically involve low mental functioning, often requiring lifetime care that also includes self-destructive behaviors. You can often see this in syndromes such as Lesch-Nyhan, Cornelia de Lange, familial dysautonomia, and extreme cases of autism. I don't think I talked about autism at all, but autism has become a growth industry, and it's almost as if, if you're born a male, you're autistic. That's kind of the way the diagnosis goes.
- 4.30 Clearly, if you were to go to Silicon Valley and test the folks there who are engineers, you're going to have, you're going to be off the scales on measures, quote, of autism. There's nothing wrong with the folks in Silicon Valley. It has to do with the scales that measure autism. It has to do with how we have so expanded that diagnosis. It's fascinating to watch the growth in this industry.
- 4.31 It began with a guy by the name of Leo Kanner, who I think it was in the late 1930s. He actually labeled autism with, I think, 14 patients. In his lifetime, maybe he saw 20. It was so rare. But then as you get into really the 1970s and 80s, there were small groups

of really dedicated individuals, particularly in California, who had kids who were bona fide autistic kids who really needed help. And they weren't getting any help.

- 4.32 And so Bernard Rimland wrote a book called *Infantile Autism*, which was the first one, I think it was in the 1980s. But it so enthused others to argue with state governments to raise particular funds of money. And once the money trail hit the screen, so to speak, the whole thing exploded. It's a diagnosis that, as I say, I think it's gone haywire. Anyway, there are some kids who had to be hospitalized with, quote, autism. And I just wanted to show you.
- 4.33 They do what is called self-stimulation. But it can often require that if you go into some of these wards, you see kids wearing helmets. Why are they wearing helmets? Because they're continually banging their heads against something in a self-destructive way. In fact, what used to be done that worked was to use punishment to stop them from punishing themselves. They used to use cattle prods. So a kid's banging his head in a back ward someplace, and you just shock him. "Hey, stop." It would disrupt that chain of events that was maintaining his behavior.
- 4.34 That, of course, now is illegal because it was misused by the staff. They started using their cattle prods to direct kids everywhere. That's why cattle move, and so would you. I mean, It's a high-current shock. But the point I'm getting at is, what may be going on in these situations is that the kid is actually hitting Chinese pressure points when they self-stimulate, and that's releasing endorphins. It's an opioid response that's going on. The way to treat that is to use opioid blockers.
- 4.35 For example, it seems to work somewhat with individuals who cut themselves a lot because that also releases endorphins. We see it, we take it to be aversive. But what is aversive for some can, in fact, be rewarding. This is a graph which shows you the degree to which you can change that kind of behaviors by giving individuals naltrexone. Naltrexone is what you use to counteract fentanyl. It counteracts opioids. You're changing the brain opioid response with this drug, which is stopping the self-destructive behavior. But understand, self-destructive behavior in kids is rare, and it's, primarily, you find it in its extreme in institutionalized children.
- 4.36 But suicide does run in families. I mean, are there any Hemingways left? I don't know. But I can name three or four that, easily, that have committed suicide. There was a study done of the Amish in the States, where four families who make up 16% of the population accounted for 73% of the suicides. We know that adoptees with depression have suicide that is three or four times greater than adoptive relatives. We know that it's more frequent in monozygotic than dizygotic twins.
- 4.37 We know also that serotonin, to a certain degree, might be involved as it's linked to impulsivity, aggression, and depression. And so there perhaps is an indirect link to suicide through depression. The major metabolite of serotonin, 5-HIAA, is very low in the CSF of suicide attempters.

- 4.38 We also are starting to learn that there are certain genes involved. There's been a lot of money that has gone to the VA in the US, the Veterans Administration, because there has been a surfeit of veterans who have attempted suicide and been successful in killing themselves. This was a study of 600,000 US military veterans, a gene localization study. They found a number of genes that were localized. A gene that actually codes for the estrogen receptor and has been implicated in depression and PTSD. A gene which codes for the dopamine receptor. A gene which encodes for synaptic plasticity. And a gene which seems to be involved in the functioning of lithium.
- 4.39 We've looked at the brains of individuals who are suicide attempters. The problem is, and why I'm just putting this there, is the nature of the problems that these brains show, which are the frontal temporal area and the amygdala in particular, areas that control emotionality, is there was no difference between column C, patients with a history of suicide attempt, and column B, patient controls. They were both depressed. If you look at depressed individuals and depressed individuals who had attempted suicide, there was no difference in the brain structure.
- 4.40 Did Edmonton lose last night? They did. I'll show you this hockey slide where players from 1967 to 2022, and it was a match cohort design. It took players from the NHL who had 50 or more fights. And there were 331 of these guys in this timespan. They're the enforcers. Every hockey team has a couple of enforcers whose job is, OK, go out there and cause a fight. They died 10 years earlier than individuals who didn't fit that role.
- 4.41 And what you can see in this small little graph is, in fact, they had significantly less tumors than their control hockey players which is the light blue. But they had significantly more suicides and drug overdoses. So clearly, there's something there, either premorbid, before they racked their brains by fighting, or in fact, they damaged their neurological mechanisms.
- 4.42 So what we've come to is suicidal prevention. And in that regard, there was a debate that went on between Tom Szasz and Ed Schneiderman. You know who Tom Szasz is, was? He actually committed suicide, but he was 90 years old. He wrote a book that, if you're in psychology and pathology, you have to read. It's called The Myth of Mental Illness. It's an old book, but his argument is, mental mind doesn't exist. I mean, if it exists, show me. Where is it? How do you measure it? There's brain disease. He'll accept that, of course. Physical disease, but mental disease? No.
- 4.43 And he was reacting to the fact that, at the time on suicide hotlines, the goal of the person talking on the phone was to find out what's your level of lethality right now. Hence, do I need to intervene right now? So they want to know, do you have the means right now? And how lethal is that means? Are you sitting in a chair and there's a gun on a table next to you? Have you been drinking? Because we know alcohol and suicide mix. Are you alone? You ask these kinds of questions.

- 4.44 What would happen at that time is, you'd trace the call so you'd know where the person is. What you'd do is, if it looked like the person was highly lethal, you'd send the police. Go intervene and stop that person. Now, Tom says that's horrible. That person is not asking you to come and to intervene. Of course, Ed Schneidman would say, "Yeah, he is. Why did they call you up? Why did they bother to call?" So it became a big issue about obligation or intrusion. They really weren't debating. They were arguing about two different things.
- 4.45 For Szasz, suicide was a right. We mental health types were interfering inappropriately. Of course, he didn't think suicide was a mental illness. Schneidman had a different approach. Ed Schneidman, along with Norm Farberow, developed the whole concept of suicide prevention. He also was a worldwide expert on Hermann Melville, by the way, and a real nice guy. He said people give warnings. They give warnings that are verbal, direct, or indirect. "I'm going to kill myself." Or indirectly, "how do you leave your assets after you're dead?"
- 4.46 If a 20-year-old comes up to you and starts talking about how leaving their assets, you know something's going on in their head. Then behavioral, direct or indirect. Behavioral direct is a trial run. Indirect is taking a long trip, unusual, spontaneous. Or it can be situational when facing certain mental or physical illnesses, where it's apparent. A person has a diagnosis of cancer or gets diagnosed with a neurological disease. Automatically, you know they're going to think of suicide. For the majority of individuals, it's a fleeting thought. But for some, it becomes a focus.
- 4.47 For Schneidman, things are cries for help, individual's emotionally troubled. See, he had this concept of what he called the congress of the mind. He said, what happens is, mentally, we think of all possibilities of things, and we kind of vote. But sometimes the voting can shift either way, like in a political body. Not in current politics, but we won't get into that. This is congress of the mind that's going on where people are continually debating, should I or shouldn't I? And so, the Schneidman's idea was, hey, if you're a professional, you come down on the side of life, and that's what you try to promote.
- 4.48 They launched the 9-8-8 Suicide Crisis Line in 2022. That's enough time for the first survey to be done, which had people complete, for almost 5,000, a psychological distress scale. 184 of 388 individuals with serious distress had heard of and used the number in this survey, and 10 of 1,000 moderately distressed had used it. The problem was, only 29% of those who had serious distress said they would use it again. So I guess it's not that functional.
- 4.49 So when you do a suicide prevention assessment, there are things you look at. Thoughts, plan, means to carry out the plan, proximity of others, hopelessness, past suicide attempts, related loss, substance abuse, and poor judgment, impulse control. What this forest plot shows, which you can see, is that, whatever you do, and you can do a number of different things, will have a positive effect in reducing suicide attempts. It's not one approach that works more than another approach. It's an approach. It's

doing something. It doesn't matter what theory you follow or whether your theoretical position is firmly planted in midair. It doesn't matter.

- 4.50** Finally, this is a comparison of effective psychopharmacotherapies of attempted suicide, the one thing that perhaps does matter. There are certain pharmacotherapies that make it more likely that an individual commit suicide. And what this graph shows is that it is particularly benzodiazepines and, surprisingly, antipsychotics and antidepressants that actually increase the risk of suicide. You're better off talking to the person than giving them a drug. The one thing that decreased risk, surprisingly, were ADHD medications and their stimulants.
- 4.51** Here's a study that is done with borderline personality disorder individuals that have a high propensity, not for suicides, but for suicide attempts. What this data just simply shows, that a treatment which is called dialectical behavior therapy is particularly efficacious. It works, I think, primarily, it's behavior therapy, but it involves behavior therapy where you focus on the suicidal behavior, and it has a preventative effect.
- 4.52** This was a study done with individuals using the Internet, who had emotional regulation disorders, non-suicidal self-injury. They had a program which was designed to teach emotional control. It had a significant effect of reducing self-destructive injury by teaching self-control. There's a panoply of things that can be done. Psychologically, you can decrease mental health issues among youth, particularly depressives and anxiety disorder. You can increase medicalization of mental illness, some antidepressants, but not most. You can work to decrease, not increase, illicit drug use.
- 4.53** What we don't want is the media to over-sensationalize, sensationalize, or normalize suicide. Family should talk about it, and the healthcare system needs to strengthen access and delivery of mental health. But that's a perpetual request. It's not going to be meant. There was a movie on Netflix a while back called 13 Reasons Why in 2017. Over a three-month period, they tracked increase in suicides. You can see that there was a slight increase after the showing of this 13 Reasons Why, in which a person—these are the reasons why an individual actually hung himself.
- 4.54** They looked at an increase in suicide rates in 10- to 19-year-olds, and it was up 12% in males, 27% in females, and there was a 27% increase in hanging. So the publication of this Netflix film actually was implicated in killing a bunch of kids. Teenagers are stupid. Get that through your heads. Now, you don't want to make a statement like that because there's some kids that are really bright. But there are reasons why we have laws related to age, and we have gotten really, really stupid about that. These kids have not gotten any brighter. In fact, they've gotten dumber. And they don't have a moral base to stand on, anyway. And then you show them stuff like this.
- 4.55** I mean, I always laugh when politicians want to reduce the voting age. Some people are arguing 16 would be appropriate. Of course, I'll get to it, but did you know that George Washington was first elected in 1756 or something like that? His first political job, to

the House of Burgesses in Virginia. And to get 309 votes, it cost him—it's in the other lecture— 144 gallons of alcohol. He bought the votes with booze, which was normal. Now, we just lower the age to produce the same effect. Anyway, sorry. Enough editorials.

- 4.56** But here's a message that's worth knowing about suicide and support. Social support is really important. They looked at individuals 19 to 20, depressed, anxious, suicidal ideation, suicidal attempts. And what was shown, that every standard deviation in social support that the individual had lowered the odds of severe depression, 47%, anxiety, 22%, and significantly reduced suicidal ideation and attempts. That's social support. That's standing up for the individual behind them, supporting them, which is different than giving them autonomy.
- 4.57** We need some steps towards a meaningful suicide strategy. It looks like MAID is made, M-A-I-D to M-A-D-E, and is part of our existence. It does serve a purpose, but it's got to be carefully, carefully monitored, and it's not currently. The politicians know that, which is why they took mental illness out of the picture for the short term. But it was a rush to judgment because right now it's very laissez-faire. You just have to convince a physician. And the reports are relative, well, they're not very detailed, but they're just actuarial. So it's a system that's going to be used more and more and more, and you have to worry about it butting up against the need to have organ donations.
- 4.58** We've already seen circumstances where, in the Veterans Administration, whatever it's called in Canada, I don't know, some of the older folks have been getting advice of, when extremely ill and thinking of MAID, the arguments have been sold on signing over organs. That's a slippery slope where there's positive value for the government to kill people. You don't want that to happen. We need better statistics and better control, period.
- 4.59** We need to view for many as a public health problem rather than a personal problem because intervening and support helps. The mantra that it is a permanent solution to a temporary problem for many really needs to be given title and followed assiduously. We need to develop and provide accessible, affordable mental health services and limit lethal means.
- 4.60** Hello. OK, so I have a bold question from what you were speaking of when it comes to suicide. It sounds like it could be a learned behavior in some cases. And I had a thought. What if suicide is an act of defense on the person?
- 4.61** Can it be a learned behavior? Absolutely. There's an old guy by the name of Nate Azrin, who—well, you can teach individuals to self-punish readily. If I'm your main caregiver and you need something from me, and the only way you're going to get it is to hit yourself, you'll hit yourself, and whatever it is. Unfortunately, that's kind of the way in which truancy often develops. It's the only way one gets attention. It's to do something which is destructive to others or self-destructive. Yeah.

- 4.62 So that self-destructive behavior is a readily learned behavior, and it's easy to, it's done every day. We learn to do all kinds of really self-destructive, stupid things because there's a payoff at the end. Reinforcement doesn't decide whether what you're doing is worth reinforcing or not. It's just there.
- 4.63 OK, so you mentioned, I want to connect some of the slides that you mentioned. One is that suicide is a permanent solution for a temporary problem.
- 4.64 Can be.
- 4.65 Can be. Another slide we had that, when you were showing the classification of the profiles, most of the profiles were for the physical illness is physical health problem. You also mentioned that the elders, especially. If we temporarily fix that suicide ideation, but the cause will not result because the physical problem for elders are not likely to be sold easily.
- 4.66 Absolutely. Yeah. There's not one answer. It depends on making a judgment. I have a friend who was diagnosed about four months ago with Lou Gehrig's disease. He's already at the point where he can barely move, can't hold anything in his hand. And six months ago, we were playing golf. So he wants to die. And when you talk to him, you want to talk about death because that's what he wants to talk about. And he would, if he could—he's not in Canada— would avail of some procedure, and he still might. Clearly a different case.
- 4.67 And the majority, from the statistics we have, it appears that the majority of suicides fit that category. It's 30-something percent. But what we're concerned with are individuals our age, your age, not my age, that have years ahead. And so for you, it might be likely, and we know it's the case. There's a study done of people who have jumped off of bridges but survived. High bridges. A very small percent ever try again. Tried it, done it, and forget it. Yeah, that's the point of looking, the fact that there are different types, and they can be classified.
- 4.68 You cited in this session here that you believe that kids are getting dumber. I'm an educator with elementary and high school-age students, and I would echo that and suggest that, and you cited that this is because there's no moral compass or bounded sets for them to really adhere to and cling on to. In my observations, as well as this, you said dumber, I've also seen an increase in what seems to be this learned helplessness and victim mentality, particularly as it pertains to mental health and how that contributes to suicidal ideation at really young ages. You cited yesterday that media is one culprit of why we are potentially seeing an increase in prevalence, particularly within younger-aged children.
- 4.69 I'm wondering if you think there's a correlation also in these increased rates or in the prevalence that we're seeing at increasingly young ages with what the school system is teaching kids, not even in content necessarily, because we won't go down that road, but just in this idea of exacerbating some of the precursors to mental health challenges,

like learned helplessness or an absence of mental fortitude and resilience, and a lowering of the standards and the bars within education that's not producing the skills children need to cope with things later in life. Could you just speak on that? What are your thoughts on that as potentially being a contributing factor?

4.70 You're right. Totally. But that's what I was implying in things that I said earlier about where society in general is going and the lack of any stabilizing force in a school system that has become so much more involved in social engineering, rather than, in fact, just teaching kids basic fundamentals. They've lost their way. Out of a kind heart, perhaps, but totally misguided because they don't have any guidance. At least, they're not listening. Yeah, and I think that's created the descriptions that you provided of what kids are like. I'll sign the editorial, that what you say is OK. It's right. I agree.

Lecture 5: The Scope of Addiction

At a glance — The scope of addiction.

- 5.1 When you're talking about addiction, we're talking about a panoply of substances and behaviors. The DSM really talks about only two things, substances and gambling. The substances—alcohol, caffeine, cannabis, hallucinogens, inhalants, opioids, sedatives, hypnotics, anxiolytics, stimulants, amphetamine, cocaine, tobacco, and unknown. If you've ever tried unknown, it's a great drug. But the fact is, there are many unknowns, and the DSM knows that. It knows that there's going to be more drugs that are going to be addictive, that are going to be addiction problems.
- 5.2 Gambling was added into the DSM-IV. They're looking at caffeine use disorder and internet gaming disorder to be included. Clearly, internet gaming disorder is going to become a rampant problem given the level of sports betting that is now in front of us. Then if we're talking about addiction, shouldn't it be anything which is uncontrolled and consequential and involves consumption, in which case it's almost an endless list and can be very individualized. But the way the DSM is set up, it'll take time to add additions, but it will grow.
- 5.3 A general definition, it's the state of being enslaved to a habit or practice or to something that is psychologically or physically habit-forming. As narcotics to such an extent that its cessation causes trauma, or simply, a fact or condition of being addicted to a particular substance, thing, or activity. That includes lots of daily activities, from the internet to video games to your handheld phone.
- 5.4 Here's a survey that was done of phones. And think of it yourself in terms of whether you're addicted to your phone or not, because I bet you are. 89% of them, this is Americans, so play the role for the minute. You can be obnoxious. OK. Americans, that probably can be edited out, right? Americans, they check their phones, 89%, within the first 10 minutes of waking up. 75% feel uneasy leaving their phones at home. 75% check their phones within five minutes of receiving a notification. 75% use their phone on a toilet. 69% have texted someone in the same room. 60% sleep with their phone.
- 5.5 60%, 57% consider themselves addicted to their phone, and 55% say they have never gone longer than 24 hours without their phone, and 47% feel a sense of panic or anxiety when their cell phone battery goes below 20%, and 46% look at their phone on a date, and 27% use their phone, text while driving. But there was actually a more recent study to show that 55% of phone users text while driving, which is six times more dangerous than drunk driving. Six times more dangerous.
- 5.6 And does it make a difference to get rid of your phone? Study was just published from Norway where they've had phones out of the classrooms for over a year. They've had a marked improvement in the performance of kids' academic performance and behavior. Albert said the following, considering the statistics I just read. "I fear the day that

technology will surpass our human interaction, and the world will have a generation of idiots."

- 5.7** There's a recent survey of, done by a political scientist and an economist, where they asked 65 domain experts. They had experts in atomic war and experts in AI. Then they also had what they called general prognosticators, 50-some members of these. They asked them two questions. The first question was, what is the likelihood of a catastrophe, a worldwide catastrophe, by 2100? So by the turn of the century. The result was, it would be a catastrophe in which at least 10% of the world would be killed or 800 million people.
- 5.8** The domain experts said that the odds are 20%. That's pretty dire. The prognosticator said 9%. Then they asked, what is the likelihood of extinction? All but 5,000 people left on the Earth. The domain experts said 6%, and the other experts said 2%. They all said the biggest risk comes from AI in terms of obliterating us. I think Albert has it right. When technology surpasses human interaction, then we truly are all idiots.
- 5.9** What does the DSM-V say about addiction? It's really specific because they say it's a chronic relapsing brain disease. They take it right into the brain, say it's a brain disease characterized by compulsive drug-seeking and use, despite harmful consequences. It is considered a brain disease because drugs change the brain, they change the structure, they change how it works.
- 5.10** Well, we know that there are a number of vicious circles involved in addiction. We know that if you're addicted, you develop cravings, you binge on a drug, you show withdrawal symptoms, and that results in increased craving. So you're caught in that vicious circle. But the definition of addiction takes it further and says that things are changing in your brain so that you get caught in a kind of a whirlpool where you become desensitized to a particular drug to a certain degree, so you're going to need more and more of that drug to produce an effect.
- 5.11** So here's the DSM's diagnosis of addiction. You have these 11 characteristics. And basically, what you're supposed to do is put a checkmark if any one of these is correct. Because what you're going to do for a diagnosis is add up the checkmarks. So is this drug taken in larger amounts and over a longer time? Is there a persistent desire to cut down? Do you spend lots of time to obtain and use the drug? Is there a strong desire and craving to use? Does use equal failure at work or school or home? Do you continue to use despite the consequences?
- 5.12** Socially, occupationally, are those activities affected? Do you use in physically hazardous situations? Do you continue to use despite knowledge? Does it develop tolerance—so you need more and more of the drug—and does it produce withdrawal symptoms? So you add up the checkmarks, and then you go to this severity scale. If one checkmark, no diagnosis. If two to three, mild substance use, four to five, moderate

substance use, six plus, severe substance use. How one would get involved clinically is towards the end of this list.

- 5.13 Let's stop for a minute and look at the natural history of drug abuse to put the whole area of addiction into a context. First, William Osler, who is a very well-known historian of medicine, he wrote the following. "The only characteristic that distinguishes man from other animals is his propensity to take drugs." He was wrong. Those are two drunk elephants. They're not dead. They come from India and are examples of elephants in a certain northern province where the locals brew a rice beer and they keep it in barrels.
- 5.14 And periodically, the elephants will break into the barrels and drink the rice beer and get drunk, so it's soused, and then crash on a local roadway. Although there have been instances where they've trampled the local residents and killed a number. And the other is a moose caught in a tree after consuming, probably, mushrooms that contained fly agaric, or magic mushrooms. And that moose thought he could fly.
- 5.15 Do you know where the word berserk comes from?
- 5.16 The berserkers, Scandinavian shirtless warriors.
- 5.17 And why were they called?
- 5.18 The shirtless-ness.
- 5.19 Yeah, it was made from a bearskin, berserkr. And that's where it comes from, going berserk. Yeah. So Osler was wrong. We have elephants and rice beer. There are apes. In Africa, a lot of the locals have learned to use certain substances by watching the monkeys. They'll dig up certain roots of trees that produce intoxicated behavior. And the locals will write about how the monkeys will go through periods of elation and then extreme withdrawal after consuming certain tubers.
- 5.20 And they've looked at how animals will select certain drugs, certain natural items, seeds, and plants to deal with wounds and healing. There's a whole literature on Siberian deer consuming hallucinogenic mushrooms and birds that eat fermented fruit and become intoxicated and fly into windows. So Osler was wrong. And animals are drug consumers just like we are.
- 5.21 So there are three reasons to use drugs. The first is to treat disorders. The ancient Greeks, Greeks used opium for depression. The early Indians used reserpine, which comes from a plant called Rauwolfia serpentina, a snake root plant, for surgeries and for serious problems, even psychotic problems. And reserpine was used as a major tranquilizer at one time for psychosis. The problem is, it could also produce the same. It's been used as a purgative to expel something from the body. And so the Egyptians would actually mix excrement with medicine, which is why today we talk about how medicine tastes like.
- 5.22 Finally, we take drugs to alter experience. There's an anthropologist by the name of Richard Rudgley, who studied Paleolithic drawings in a number of Europe caves and

compared some of the designs that were drawn to what people will draw, if they're asked to draw a design, if they're on a peyote trip or in some form of some hallucinogens. And it's the same. The same constructs are produced, which led him to believe that the use of hallucinogenic mushrooms in these Paleolithic drawings, probably the use of the mushrooms predates history.

- 5.23 The opium poppy has been cultivated, at least from the fourth millennium BC. The Greek historian Herodotus writes about cannabis-intoxicated Scythians who would meet in groups and consume cannabis and Twinkies. No. Some other kind of munchie, obviously, whatever Scythians munched on in the fifth century. The early Egyptians were users of beer, wine, opium, mandric root. Incas of Peru chewed coca leaves, and finally, the Aztecs and hallucinogenic mushrooms.
- 5.24 I was going to bring you that, I forgot it, a statement from the Florentine Codex, which would have been around late 1600s, I suppose, 1700s in Mexico City, which describes Aztecs and hallucinogenic mushrooms. The nature of the response is these people would consume the hallucinogens in groups, but then separate individually. They then described different experiences. I think I mentioned this last time.
- 5.25 What was unique about the experiences is that I think about three-fourths of them were quite negative. They would see themselves being sacrificed or being killed in some way, whereas one quarter, at least, would see themselves being rewarded in some way, and something glorious happening, illustrating an extreme divergency of response. The point being, with hallucinogens, and with most drugs, depending on dosage, it's what we bring to the situation which is really important. I'll talk a little bit about marijuana research and show that later.
- 5.26 This is just a drawing that appears on a cave in a tomb in Egypt, and they're drunken revelers being removed. And Egypt was the first governmental body to actually pass drunkenness laws. So they've been around for a while. This shows the fall and rise of alcohol in the US and England. And today, we consume something like two and a half gallons, US gallons, of alcohol per person. But what's unique is the role alcohol has played in our development.
- 5.27 I told you the story about George Washington. Did you know that the Mayflower landed where? It's where—Plymouth Rock. They weren't supposed to land there. They were supposed to land at the mouth of Hudson River, which is New York City. They got tossed because they were drinking all the beer. And the sailors who had the ship wanted to keep the beer for themselves. So it was a beer fight, and the pilgrims got dumped in Massachusetts.
- 5.28 There's all kinds of history of alcohol and how it's kind of controlled the directionality of society. There's always been exclusions to drug use. Every society has excluded one type of drug or another, or advocated that only certain individuals could use a drug. Typically, it was people of authority who could use hallucinogens, and the most of

others could not. Because it was only thought that, for example, in some societies, that holy people can, in fact, hallucinate appropriately. But let me show you how we've changed.

- 5.29 From the 1908 Sears catalog, you can order the following. You can order a soothing syrup which includes opium, and it was designed to soothe babies. Babies were given a tincture of opium in this syrup. There's a paregoric, which is used for digestive problems today. There you are buying cigarettes at your bedside in the hospital. Stuff that is anathema today because this is what we tend to see. It's the case now that each cigarette has a warning, doesn't it? So you can't get away from it.
- 5.30 But what has happened is that the government has been successful. Society has been successful in cutting the use of cigarettes, even though, I'll later show, they are dopaminergically the most addictive. So it can be dealt with on a societal level. These are graphs that basically show the fact that smoking prevalence between 2011 and 2022 for adults ages 18 to 24, and the second one is adults, 23 to 29. And the level of smoking has gone down in just those few years.
- 5.31 So let's talk about some general issues that exist. And that is that we tend to have bursting prisons because we cannot decide whether drug use is a legal issue or is a health issue. We are clearly caught between those two views and have accepted the position that it's both. 55% currently of drug arrests are for possession. You'd get the impression that that's not the case, but it is. In the US, over a million individuals are in jail or on parole because of drug crimes. And crime is subsidized because, the last figure I saw, 87% of cash generated by organized crime is from drug money. The net effect of that is to increase the price of drugs.
- 5.32 Even to the extent that the DEA puts out charts like this, which shows how the country of Mexico is divided by its cartels. The cartels, their major source of income, it's drugs, although human smuggling is starting to rival that. But it's exceedingly violent. In their last election, how many people running for office were murdered? Do you remember? Like 17, something like that, 15, 17, of individuals running for office were murdered. Death rate is exceedingly high.
- 5.33 And why all of this becomes more and more of an issue is that more and more people are dying from drugs. The first slide basically shows how that level has increased dramatically. Currently, over 107,000 individuals died in 2022. The slide next to it basically shows why. That's because there's been this dramatic increase in the use of fentanyl. Fentanyl is a manufactured narcotic that was made as an anesthetic designed in 1950s.
- 5.34 This is a very interesting slide because what it shows is the relative effect of heroin. The size of the box is what you need to get a typical effect versus a drug like fentanyl, which is one-fiftieth, takes one-fiftieth of the drug to produce the same effect as heroin. Now, the problem with fentanyl is it's being replaced. It's being replaced by a drug called iso,

or isotornitazene, which is one-250th of the size in terms of producing the effect. It's far easier to produce these drugs that are small, easily transported, easily smuggled, and actually easily made. And they're very, very powerful.

- 5.35** And so, you see maps like this that show how, in this case, fentanyl goes into the United States, being manufactured in India or China. But that's not critical. Provided you get the basic chemicals, you can do it in your basement. But they're seen as being shipped primarily to Mexico and then infiltrated into the US. But there are a number of routes that occur, as well as the machines used to make the drugs.
- 5.36** So currently, the people who are dealing with the drug problem are wedded to the idea of harm avoidance. And harm avoidance means you want to do everything you can to minimize the negative effects of a drug. In this case, the effect of the drug is death. So we know that a drug like naloxone inhibits fentanyl and opioids. It actually, in some people, also inhibits the high of alcohol. It's because it operates on the dopaminergic system.
- 5.37** In the States, you can get naloxone at any local pharmacy. It comes in two varieties, a nasal spray and an injection. If you're an addict, you can get fentanyl strips that test the drug you're using and identify whether fentanyl has been added. Because it is the case, the study was done in New York City, that they looked at the drugs that the street users were using, and they asked the individuals, "Are you using fentanyl?" It turns out that only 13% said yes, that they knew it. The majority were using fentanyl and didn't know it. It was mixed in their drugs.
- 5.38** In this case, you're dealing with people who, their quality of life is less than ideal. Ideal is a score of 100. Their general health, body pain, social functioning, mental health, vitality, role limitations due to mental health, the best they can do is 40% out of 100%. In any kind of objective situations, they're all failing, and their lives are poor to miserable. Nevertheless, funding has been increased, and you can see from this graph that larger amounts are slowly going to treatment, and treatment is critical.
- 5.39** There are drugs. I can't pronounce it. But there is a drug that is long-acting and can work for weeks to counteract the highs of opioids. Trouble is, it's not that easily available. If it were easily available, it would be quite therapeutic for a lot of individuals, provided they take it.
- 5.40** The other problems with negative aspects of drug laws is violence and crimes are increased. As I said before, if they're illegal, things become more expensive. And addicts henceforth live in a world of crime, often required to commit crime to get money to buy drugs. So it's a vicious circle that they live in. The poor are victimized. The drug use often occurred in modest environments. And for some, drug traffickers are the people who have the cars, the money, etc. And they become status symbols.
- 5.41** On the most part, the law does not get respected. It undermines respect the total legal system. And that's why if you're doing certain aspects of harm avoidance, the laws can

prevent you from doing it. At the very least, you ask the police to be benign. So in the States, and in Canada now, we have recreational cannabis legalization. In the States, there's 24 states that would be equivalent to what Canada has, although each province is different in how they control the sale of cannabis.

- 5.42 There was a study done with almost 900,000 high school students. And what it showed is that in those 24 states where cannabis could be bought, there was actually a drop in sales to high school students. And it was associated with a 28% higher odds of zero use, except for those who had already been users, where the odds increased 26%. You get maps like this. Now, remember, cannabis use is a schedule one prohibited drug in the US, federally, except for those 24 states, where you can buy it, but you can't necessarily take it to the next state. If a federal officer saw you, they might have to arrest you. So legitimate use is also obviated.
- 5.43 There are many treatments. We talked about with depression, how psilocybin is now being used to treat depression. And there are some studies that have been very optimistic, and we'll see whether that pans out into more usage. The drug laws also promote hypocrisy, and certainly when you see a patchwork like I just showed you, I mean, how hypocritical can you get? You can walk across the state line and use a drug or not.
- 5.44 What's been happening in Canada and the States to a large degree was based on what people thought was happening in Portugal. And because Portugal produced statistics like this, which showed a very low death rate from taking drugs, so Oregon and BC, BC asked to decriminalize the use of cocaine, opioids, methamphetamine, and ecstasy, and the federal government agreed. BC is now asking the federal government to go back. What happened, is happening, in BC, is it Hastings Street? Is that the street that used to be the center of drug use in Vancouver?
- 5.45 East Vancouver, Hastings, yes.
- 5.46 I was there years ago, and it was the classic picture of drug use. And what the people who were working there were struggling to do is find rooms for people, places where they could crash and live. And it was quite circumscribed. Because you walk a few blocks away and it wasn't a problem. Since BC has passed this law, it's everywhere. It spread. So you walk into a library in a small town in BC, and it might be somebody's stoned there. And so there's been a popular uprising against the situation. And the folks, the government in BC that had been so liberal is, in fact, now saying, "Perhaps we went too far."
- 5.47 Oregon, that had, in fact, passed a decriminalized law, had the same thing happen. It became a much more universal behavior. It's like, what you need is a medical approach but not acceptance. And that's tricky because what's happened is acceptance. Everyone got a green light, and that's turned out to be a blunder. And so things have to change in some way. So Oregon is currently considering repealing their law.

- 5.48 And then there's Toronto, which wants to do what BC did. They should check with where BC is at right now. Because right now, they wouldn't do what they did. They know it's politically—it might be OK for the addict, but it's not OK for society, because it's not a model of society that you want. You don't want to feature people who are dependent. They're ill, but they're ill in a way that is hard to integrate into your society. And so it's not going to work, and it's not working.
- 5.49 Everything we have is how you look at a disorder and how you label it. For example, if you look at the relationship of what alcohol does and what opium does, does it produce cell death in the gastrointestinal system? Alcohol does, yeah. Opium doesn't. Does it produce cell death in the central nervous system? Yeah, alcohol does. Opium doesn't. Does it produce neurobiological brain defects? Alcohol does. Opium doesn't. Does it produce inevitable physical deterioration? Alcohol does, and opioid, opium doesn't. And I say, what about chocolate? Chocolate directly affects cannaboid receptors in the brain.
- 5.50 If you're outlawing drugs, you have to consider these kinds of things. One can learn not to use certain substances and learn to use other substances and not to overuse substances. We'll see later that it, to learn that, you have to know yourself because some of us are much more likely to be addicted than others. There was a massive study done with US servicemen coming back from Vietnam. And many, I can't remember the percentage, but many, they all had to have physical exams. And many of them were addicted to heroin. Or they'd show high levels of heroin in their blood.
- 5.51 They were followed by the VA, and only a very small percentage of them went on to be heroin abusers, even though they'd been using heroin and opioids in Vietnam. Because you're in the middle of a horrendous situation and you want to be spaced out, but coming back and changing circumstances stopped it. So what that says is maybe they didn't have a biological potential to become truly addicted, that they benefited from the effect of the drug, but could let it go. And that's the case with many of us, with many kinds of substances.
- 5.52 And so there may be things about people, and I'll get into that, that make them more likely to be addicted. But we do want to keep drugs from kids. We don't want to create more problems. And we know that there are real and negative consequences to drug use. This is a graph which shows the increase for males and females in terms of children who lost a parent to drug overdose. We know that kids who are orphaned don't do that well or who are stuck with a single parent. Foster care involvement of children of opioid use parents, particularly the individuals of the youngest age, ages 4 to 8, are apt to end up in foster care if their parents are opioid users.
- 5.53 This is a slide which shows the mental health, cognition, academic performance, and delinquency of non-disordered cannabis use. So these are individuals who are non-users. That's the dark blue. The gray is non-disordered cannabis users versus the light gray are those with cannabis use disorder. You can see that the cannabis use disorder is

the worst in terms of mental health, but that is approached by just simple cannabis users. They have far more mental health problems than do non-users, and yet it's government-sponsored.

- 5.54** So drugs have an effect. Their cognition is worse, their academic performance is worse, and their delinquency is higher. Alcohol, another legal substance, we've seen in recent years an increase in deaths caused by alcohol for males and females. Here's a map in the US of alcohol-related deaths, deaths attributable to excessive alcohol use among age, individuals aged 20 to 64. It's in those areas sparsely populated.
- 5.55** What else do you know about those areas that would increase the statistics dramatically? It's the same as suicide, wasn't it? It's a sparse population, and it's heavily populated by aboriginals, American Indians, and they have a very high suicide rate and a very high alcohol consumption. As I showed you that slide on stress, they also have a very high stress level. Just to consider alcohol, which is legal. It's the third most prevalent health problem.
- 5.56** This is the data I used whenever I'd write an alcohol grant. Because you know when you write a grant, you have to say how bad your problem is? So you search for the worst statistics you can find anywhere. So alcohol is the third and most prevalent health problem. It's a primary cause of emergency room visits, US. In Canada, we go to the emergency room because we can't find a doctor. But that's another reason.
- 5.57** It's involved in 39% of traffic accidents. Thank you, Mothers Against Drunk Driving, for reducing that number, which was 50%. Causes one-third of general hospital admissions, 20% of male deaths, 15% of female deaths are attributable to alcohol. It's involved in 50% of violent crime. 50% of rape victims were intoxicated at the time of the crime. I get shit for saying that in mixed audiences because they say I'm blaming the woman. But it's a fact. It's a fact of how one can be a victim of crime because, what alcohol is doing, it's dampening those areas of the brain that allow you to think evaluatively of what's going on in situations. And alcohol blurs that.
- 5.58** A study we did, we looked at 11 countries, 26 studies, almost 10,000 cases. And in those, 61% of offenders in violent crimes had been drinking, and it was mentioned in their files. And the percentage of victims that had been drinking overall was 45%.
- 5.59** Medical consequences of alcohol use, anemia, arthritis, convulsions, diabetes, high cholesterol, hypertension, coronary heart disease, gastro diseases. Alcohol is a primary factor in a lot of physical disorders, maybe even in difficulty talking. Let me just point out that the effect of any drug is a function of dosage, and not only dosage, but route of administration. So there's some basic pharmacokinetics that one always has to consider. What is the street addict doing now? You used to find places where there were just a lot of needles left because they were injecting a drug.
- 5.60** But now, if you're using fentanyl, which is far more powerful, you smoke it. You're going to find little pieces of tinfoil on which they've smoked the small bits of drug, and

it controls the speed at which it gets into the bloodstream. We know that one's age, gender, diet, environment, body weight, health, drug interactions are all important and affect what a specific drug does to an individual.

- 5.61** You also have to be concerned with how it is absorbed, how it gets distributed, how it gets metabolized, and how it gets excreted. If you don't consider pharmacokinetics in drug effects, you don't really understand drugs. Let me show you how that's important. In this case, we're talking about dose. In the 1970s, a common belief was that cannabis produced brain damage. We looked at that data and said, no, something's wrong. Because all of the data came from street people, which showed the fact, A, that they had mixed many different drugs and thus were living a very difficult lifestyle, and that in itself could explain their neurological findings.
- 5.62** We wanted to find an environment where cannabis use was frequent and acceptable, kind of acceptable. So we went to Jamaica. It was also winter in Montreal. We actually went to Jamaica twice. You need to do that, twice. The first time around the hills of Montego Bay, we tested people who were cannabis users. And by the way, half the population there was stoned, was using the drug on a regular basis. So it was the right place to be.
- 5.63** And so we tested people, and we weren't getting any effects. We had some control effects, like the degree to which you're educated would affect performance. And that was turning out to be true. But in terms of the neuropsychological stuff that we were looking at, we weren't finding anything. That's why we did another study in Trench Town, in Kingston. That required political machinations with local Rastafarian chieftains, but it worked. The woman who did that is just brilliant, later became a dean at Simon Fraser.
- 5.64** We basically found the same thing, that ganja use could pretty much do whatever you wanted it to do. We said, well, maybe the drug they're using isn't potent. So we smuggled the drug out of Jamaica because I had an import license for Canada. So I could import the drug, but I couldn't export. Anyway, we just mailed it and gave it to the, to Ottawa to analyze it, and the average THC content was 6.7. THC is the active ingredient in marijuana, right? Well, heck, yes. That was equal to hashish at the time. So it was powerful of what was going on.
- 5.65** And we didn't find anything, other than the drug worked for Rastafarians in terms of how they wanted it to work. If you wanted to feel healthy, it worked. If you wanted to be relaxed, it worked. If you wanted to be active, it worked. It depended what you wanted. It was a powerful drug, but the point I'm making is it pales in significance to cannabis today. Cannabis today is a very different drug in terms of dosage. There isn't a control dose that says you can't go over this in THC. And the government's very lax.
- 5.66** They do it with alcohol. Alcohol is limited in its content. Would you drink a beer that has 20% alcohol? It wouldn't take much of a beer to put you under. Anyway, because

this data shows that in terms of usage of today's cannabis, where you're getting THC contents that go up to 20, 30%, I mean, that's a wicked drug.

5.67 The MRIs show differences in the frontal lobes in individuals relative to the kind of dosage that they use, so it does have a brain effect. And then I looked at data from a poison reporting center. And this was all edibles. This is the growth in, over years and by age, in reports to a poison control center from overdosing on cannabis. And it was all cannabis edibles that were highly dosed.

Lecture 6: Breaking the Cycle

At a glance — Breaking the cycle.

- 6.1 This report shows global trends from a report in 2023. It shows that the people who use drugs were almost 300 million worldwide, which is a 23% increase from previous years. The most used drug worldwide is cannabis, followed by opioids, followed by amphetamines, followed by cocaine, followed by ecstasy. The figure also shows you what drugs are being cultivated with a 35% increase in coca leaves and a 28% in opium poppies.
- 6.2 The next figure shows the relationship in drug use between males and females. For the most part, males use far more drugs than females. In terms of countries that use drugs, it's basically America, Europe, and interestingly, Oceania. Just some more statistics. This is data in the US of people who used illegal drugs in 2021. It shows that marijuana was the most illegally used drug, although it's not illegal in most states, as I showed you, at least in 23 states, followed by pain relievers, hallucinogens, tranquilizers, cocaine, stimulants, methamphetamine, all used at a significantly lesser amount.
- 6.3 There's a tremendous cost involved, which is what this figure shows you, where if you're advocating for drug interventions, you look at data like this because it draws costs from everywhere, and they come up with costs in the billions. Finally, your statistic on drug use disorders by age, and it shows that most people who use drugs the heaviest are in their 20s. The type of drug used are illicit drugs and/or alcohol and opioids.
- 6.4 There are certain mental health problems that use more drugs than others, with the most likely category being intermittent explosive disorder. Now, if you haven't got a clue what that means—and I didn't, I had to look it up—intermittent explosive disorder is recurrent outbursts, verbal, physical tantrums, arguments, fights, twice weekly for three months. Grossly out of proportion to provocation, not premeditated, causes distress, and at least six years old. I guess it begins early.
- 6.5 But that's followed by ADHD. That's one problem with ADHD because the automatic treatment is with a psychostimulant, methylphenidate. The problem is that it also has been associated with a lot of drug use. We did a study, in fact, where we looked at adult methylphenidate users who had prescriptions for ADHD and found that up to, I think, three-fourths of them were selling the drugs. So even though they had a prescription, it was being utilized more broadly, became a source of income.
- 6.6 In particular, when we're looking at drug abuse, we're looking at college students, because if you're a university student— and you're not in a tent right now, that's a joke — 46% had a DSM disorder, or maybe that's the folks in a tent. 21% have an alcohol use disorder, 14% nicotine dependent, 5% other drugs. So drug abuse is a large percentage, half of that or more of that, 46%.

- 6.7** The lifetime prevalence of an alcohol use disorder is 10%. For males, it's almost 14%, for females, it's half of that. For Native Americans, Native Canadians, it's almost 19%. Peak age is teens, 20s, highly comorbid with mood, anxiety, and personality disorders, and it's unlikely ever to be treated. Only 8% of abusers are ever treated, and only 38% of those who have severe alcohol problems. We can look at past month alcohol use by age, and you can see how it really peaks in those 18 to 25 and 26 and older.
- 6.8** What factors control drug-taking behavior? They're not necessarily the same factors. There are initiation factors, learning to use a drug. There are individual factors. Who you are matters in terms of whether you continue to use. Then there are models of maintenance. What keeps drug use going? That's affected by different biological systems. That's determined by what the drug does for you, which can be physical and can be psychological.
- 6.9** We've had a strategy where we felt that there are multiple and often very discrete ideologies, and that to understand them, a high-risk approach is required. We thought that what would really be important is to look at how that person responds to the drug is going to determine whether the drug will be a problem. This is called a drug challenge. We've done a series of drug challenges, experiments.
- 6.10** But we also said that what we want to do is look at people who are at risk for developing a problem. Don't have it now, but they will. We're betting they will. We look at contemporaneous, we do what's called high-risk research. We find people who are at risk to develop a problem. We treat them with it. We give them a drug, and we see how they react. That's a contemporaneous recording of events. Because if you study alcoholics, for example, or any drug-dependent individual, you're looking at the explosion, not the causal mechanism.
- 6.11** Anything you learn about their brain or their personality could be because they used drugs, not because they started using drugs, or not because the drugs became addictive. Studying people who have a problem, as I say, you're studying the explosion, not the triggering mechanism. And so if you wanted to understand etiology or cause, it's the triggering mechanism that you want to understand. Because if you do that, you can also see how some people don't develop the problem. They should, but they don't. Why?
- 6.12** You can look at the fact that being at risk for a problem might mean you develop other problems, the heterogeneity of abnormal outcomes. You can see how some things occur at certain ages. The genetics of schizophrenia, for example, are strong. They're there. These individuals should have problems really early. Some of them don't. Lots of them don't. It's not until the late teens, early 20s, that you start seeing symptoms. So there's a development that goes on. It allows for the assessment of changes over time within individuals because they do occur.

- 6.13** People have disorders often in cyclical fashion, and the disorders change. You can understand feedback mechanisms and circular prospects in causation. We're looking at alcohol and we're looking at who's at risk. Well, we know people with a certain family history of alcoholism are at risk. Certain personality traits put you at risk. Certain diagnoses, ADH, CD, ASPD, put you at risk. Anxiety, depressive disorders, do the same. Early alcohol use, certain occupations, premature adolescence, social failure, etc.
- 6.14** Here's what we look at, initially. We wanted to take what we thought was the strongest case. What we're looking at is the diagnosis of alcohol in a family. You can see how it's male-inherited, as males are the dark boxes, and they collectivize. But the subjects we study are those three guys down from the bottom that are at risk, but they don't have an alcohol problem. But if you look at their pedigree, you can say, gee, why not? Everybody else who's male in that family has a problem.
- 6.15** What we know about individuals who have this history of paternal alcoholism, and alcoholism is far more inherited in males than it is in females; doesn't mean it's not inherited in females, but you don't find it like this. And these are done from cross-fostering studies. These sons of alcoholics have particular psychophysiological findings. They're autonomically hyperreactive. They have certain cognitive deficits, tend to do poor in school, and on certain cognitive tests tend to have had history of conduct disorder, hyperactivity, and they have interesting and discrete reactions to alcohol, which is what I'll show you.
- 6.16** I'm going to do a number of experimental things we've gone through, and this is just one set of studies. The first thing we do is get a baseline sober heart rate. We measured heart rate for a number of reasons. Then we do an alcohol challenge. We get them drunk, and we see how they get drunk and how it's different from others, what it does for them. We test their heart rate, of course, because that's one of the measures we use. We look at cognitive functioning, learning, drinking behavior, brain activity. If you can think of a variable, we'll look at it.
- 6.17** We look at heart rate response to alcohol intoxication because we know that heart rate beats to rewards. Its change reflects all kinds of things, but this is the one factor we're looking at. If alcohol is more rewarding, heart rate should go up, and it does. These are family history negative, the first set of groups for sober individuals versus alcohol-intoxicated. The second is another population we looked at, who are also at risk, anxiety-sensitive individuals who are also at risk for alcohol disorders. And then the third group are those with this multifamily history. You can see that we get an increased heart response significant in the latter two groups.
- 6.18** Here's what's fascinating for those two groups, is if we put them in a situation where we're again measuring change in heart rate and we shock them, there's a countdown, 10, 9, 8, 7, 6, down to 1, and then they get an electric shock. Individuals, drunk and not drunk, but control subjects, react. Their heart rate goes up for that, too. But not if the two groups, the anxiety-sensitive and the multifamily history boys, their heart rate

actually goes down. It's negatively reinforcing and rewarding. They don't react to aversiveness when intoxicated. So intoxication, you can see, is going to minimize aversive events as they occur.

- 6.19** In terms of the general increase in heart rate, we looked at a number of different groups. We looked at individuals who had no family history of alcohol. That's the first two. We looked at individuals who have one generation of male family history, and they have no reaction. Then we looked at women and men from multi-generational, and that's the next two. The women had a decrease, and the men had a dramatic increase. The last group is a group of alcoholics that we got intoxicated.
- 6.20** Furthermore, we tested them in a number of situations. This was a test of impulse control, and it turned out that the high heart rate group, when intoxicated, actually was able to control their impulses. And in intoxicated, they even, they showed more risk-taking. We wanted to see if this occurred early, so we were using 11- and 12-year-old boys. All of our other subjects are legal, so they can drink. But these are young kids from this multi-generational history.
- 6.21** We showed that, to begin with, they do show cognitive problems. They do poor in terms of word fluency, the Wisconsin card sort, and self-order pointing, all frontal lobe measures. This was a study in which we showed that if we had alcoholics learn a list of words, got them drunk, had them come in the next day to do a memory test, that if they had gotten drunk, they remember more of the emotional positive words than if they had not gotten drunk. For some reason, they were able to focus, pay attention to positive things, which in fact was facilitory.
- 6.22** Finally, just a few other things. They will tell you that when they drink, they really feel good. That's all the first graph shows you, is that when intoxicated, they really feel good, and they feel good for a longer period of time. The next is the blood alcohol curve versus normals. For most people, if we get the normal subject intoxicated, they peak rather quickly and then drop and actually go below baseline. These guys, they stay high for a long period of time, much longer, in this case, over an hour.
- 6.23** We show that they self-rate more positive responses, they drink more, they remember more words to learn before drinking. They have increased plasma endorphin levels. Their heart rate response is blocked by naltrexone. It's a real positive experience. It's a positive experience that is shown emotionally, cognitively, and neurologically. That whole system is operating. We demonstrated that we can block everything if we give them naltrexone. So what does that tell you? For these guys, alcohol is directly affecting their dopaminergic system.
- 6.24** And we actually showed that. We did a PET scan and showed that, with alcohol, in particular these guys, you promote the release of dopamine in the nucleus accumbens. That's just another view of a different group of subjects showing that release of dopamine. But it's not just alcohol. What this graph shows is, see the degree of effect

of the release of dopamine for various drugs, for amphetamine, for oral amphetamine, for nicotine, for alcohol, for ketamine, and a little bit for THC, but not that much.

- 6.25** Now, if I were to put morphine and opioids and fentanyl here, it would look the same, only more so, because they really hit the dopaminergic system. That system, there's actually two dopaminergic systems, but they both energize the prefrontal cortex and the cingulate cortex, which wraps around. But it's, it hits areas that are signals of action, are signals that reward is possible. It's also that punishment as possible, but it can be very pleasurable. These emanate in the nucleus accumbens in both dopaminergic systems. As I say, they implicate the frontal lobes.
- 6.26** Now, what is interesting is that, if you take that baseline as being a normal response, this is a degree to which normal responses occurs to most of these drugs. The other one was individuals who are addicted. So there's a difference in the brain response of dopamine of individuals who are addicted versus those who are not. For most of you, you could use, not suggesting, some illegal substance, and it would be nothing, although I'm not sure of the new drugs that are really powerful.
- 6.27** There was just a recent study, this was just published, where they looked at genes that might be involved in a genome-wide association study of over a million drug users. The drug that you can't see, which is, I can't name that. That one right there. That's the one that was most prevalent in all of the one million drug users they looked at. Again, it's the DRD2 gene. It's a gene which is involved in dopamine receptor.
- 6.28** There is, or we can say, unequivocally, that most drug users do represent a physiological susceptibility which is related to the dopaminergic system, which is why certain drugs "take control." And if you're going to treat them, you have a choice. And what we've done is find a drug which is more acceptable. And so what is our treatment for a lot of these folks? We use methadone or buprenorphine, I can't pronounce it, which is long-acting. But it's an addictive substance, too. It affects the same system, but it's legal. It's medical.
- 6.29** We use a drug which does the same thing, not quite to the same degree. And that's our treatment. But what we're doing is recognizing that we have a problem. What is your choice? Choice is abstinence or you capitulate and say, "Let's go with something which is legal, which we can do in a clean way," which is acceptable. I know a lot of people who were dependent on certain drugs, typically opium, were physicians, doing well, had a source of drugs. So some things can be accepted without worrying about side effects, which is, in fact, produced by how we approach it.
- 6.30** What we're finding now is the same kind of thing is happening in gamblers, where you find an underactivity of the mesolimbic reward system. And what is that? That's the dopaminergic system. We're going to find increasingly more problems as dopamine gets associated with certain feelings of reward in individuals who don't have much reward, have to find it elsewhere.

- 6.31** I mean, the current opioid crisis really had its impetus in today's medicine. Why did Purdue Pharma get fined billions of dollars? Because the ad agency itself got fined multiple billions of dollars. Well, they were pushing OxyContin. I had a house in Florida, and next to the post office, there was a small building. It was a nursery school at one point. And I went to the post office once, and that whole street was jammed with hundreds of cars, and there was a lineup of people down a block. One physician had moved into that office and was writing prescriptions for OxyContin, and it became a source.
- 6.32** The legal system did it. The medical system did it. It was not unusual to write scripts for OxyContin for individuals who would have a wisdom tooth removed. They developed addiction, some of them, because, again, the effect on the dopaminergic system. And we can see a similar effect occurring to a slightly lesser degree in gambling. There's been a recent 75% growth in gambling problems.
- 6.33** If you look at pathological gamblers, 26% were divorced, 34% lost their job, 50% stole money, 18% got arrested, 6% had suicidal thoughts, and they even developed tolerance. To get a diagnosis of pathological gambling, you need five more of the following: preoccupation with gambling; increasing amounts of money needed to achieve desired excitement; repeated efforts to control and cut down; restless or irritable when attempting to cut down; gambles as a way of escaping; after losing money, often returns to get even; lies to family members, et cetera, has committed illegal acts to finance gambling.
- 6.34** What did that guy who worked for Ohtani, gambled away 5 million? You don't watch baseball? Has committed illegal acts, jeopardized or lost significant relationships. And this gambling behavior is not better accounted for by a manic episode. So pathological gambling has similarities to—and the DSM can justify it as being included because, like drugs, we're seeing the similar brain effects in terms of etiology.
- 6.35** Not all drugs have dependence liability, caffeine, antidepressants, and they don't affect the mesolimbic dopamine system. But you can look at these other drugs and see the estimated prevalence of dependence. Like in nicotine, it's about a third who tried. Heroin, 23%, cocaine, 17%, alcohol, 15%, stimulants, 11%, cannabis, 9%, psychedelics, 5%. We started studying people with genetic predisposition. But in understanding drugs, that gets you to neuroadaptation and changes in the brain and sensitivity that occur.
- 6.36** It is the case that use of the drug leads to decreased functioning of this dopaminergic system, hence the need to use more drugs. To understand drug abuse, you look at adoption studies, twin studies, linkage studies, flying paper. If we just look at twins, dizygotic versus monozygotic for abuse and for dependence.
- 6.37** For cannabis, you can see that there is definitely a twin effect, both for abuse and dependence. A big twin effect in sedatives, for twin, for dizygotic, monozygotics. Kind

of a lesser one for cocaine, which means that cocaine is going to affect some people who don't have a heritable predisposition. Stimulants, again, not so much for dependence. Hallucinogens, a big effect by heritability.

- 6.38** Genes that are implicated, I already showed you that dopaminergic gene for alcohol, but there's the ADH, alcohol dehydrogenase gene, dehydrogenase gene, a serotonin gene. There's dopamine, GABA receptor gene. A number of genes have been determined. There've also been a lot of studies done that are called reactivity studies, where basically what you study are individuals who show this biological reactivity to the drug, and you look at their brains and how they respond to a particular drug.
- 6.39** What you can see in this meta-analysis is that they've been looked at in many different ways. But the fundamental finding is that there's no finding. No, that is the fundamental finding. The problem is, individuals with bipolar disorder and alcohol use disorder, this was a study where they looked at individuals who had either one of these problems or both and showed a differential performance in terms of differences in the right side of the brain, in terms of the right frontal gyrus, insula area.
- 6.40** And it's very distinct. So these are individuals who have alcohol use disorder plus bipolar disorder. Their brains are very different from individuals who just have AUD, bipolar disorder, or are normal. You have to be careful about who you're studying.
- 6.41** There are many different motivational systems that affect how we respond to drugs. There are environmental triggers, genetically influenced susceptibility, and differential drug responses that are reinforced. I want to show you a set of studies that illustrate that point. We asked the question, how do anxiety-sensitive and sensation-seeking individuals differ from each other in their neural responses to an alcohol challenge and to stress manipulations? And are there gender differences?
- 6.42** So here's the experiment. It's a scanning experiment, double-blind, placebo-controlled. So we give individuals a drink of alcohol or placebo. The alcohol is designed to produce a blood alcohol level of between 0.08 and 0.10, but it actually fluctuates a bit more. So they're legally drunk. In Ontario, they're really drunk. So this subject had 11 anxiety-sensitive males and 11 sensation-seeking males, and the same for women. And after a bit of rest so that we knew the alcohol was being absorbed, after 15 minutes, we stuck them in a scanner, which is probably a place you don't want to be.
- 6.43** And then we stressed them. So it's not enough that you're just there. We're going to make it unpleasant. And one way we made them unpleasant is, while they're in the scanner, this is a common procedure, we showed them these Karolinska faces; and these faces are all designed to elicit different mental responses. And in particular, we showed them all the faces, but what we were focusing on was faces that would show emotionality and fear, because that had been shown to target anxiety.
- 6.44** Then we gave them also a stress manipulation of a different kind. It's called the Montreal Stress Task, and it involves math and social pressure. So they get a math

problem, and they're timed, and they have to solve the math problem. In the meantime, the experimenter, who is a young woman, is standing there, like a first-grade teacher, in your face, get it right, you can do it approach. She's supposed to be the stressor. And by the way, we found it makes a big difference in terms of studies you do and whether you have the quality of the experimenter. They make a tremendous difference in studies, and nobody talks about it.

- 6.45** We used to do studies with tryptophan depletion, producing depression. It was also called the "upchuck experiment" because I tried it, and it's effective. It's just terrible to drink. So it's got this reputation of leading to nausea. So you don't want people barfing in your lab all over the place, although that's why you have graduate students. Anyway, we found that there are two very effective experimenters. One is, we had an ex-football player who played college ball and was a graduate student. People didn't throw up when he was around. When he was the experimenter, you held on. Or an attractive woman. Powerful forces, socially, to inhibit nausea.
- 6.46** Anyway, so we're bugging these people. You can see here, this is in anxiety-sensitive individuals. In the placebo condition, look at how their brain lights up and how the hippocampus, the amygdala, and that fiber tract, the caudate, are all hyperactive. They're responding to that anxiety. They're showing an anxious brain. Look what happens when they're drunk. It's gone. It doesn't bother them. They're not anxious. That's to the faces. This is sensation-seeking individuals under placebo and while drunk. In neither case, didn't matter. No effect.
- 6.47** So clearly, if you're looking at what a drug does for a person, for that anxiety-sensitive individual, alcohol is going to be a problem because it's going to dampen their emotional responses that bother them. Here this shows for the left hippocampal response to fearful faces, the left amygdala response to neutral faces under alcohol and placebo. We didn't find anything which jumped off the charts in terms of response to that math stress test, although males reacted dramatically in terms of being stressed, but women didn't.
- 6.48** So either they're smarter and far better at math or I ain't going to respond to another woman. No way. But they didn't. Anyway, whatever it is, it shows control in the non-susceptibility. We did find differences in looking at brains in terms of how they responded to alcohol. Sensation-seeking females had far more activation of the brain than did anxiety-sensitive females, period.
- 6.49** That correlates, by the way, with another study we did with female prisoners. We went into a jail and looked at female and tested female prisoners on a bunch of scales. And we found that, on the whole, incarcerated females scored high in sensation-seeking sensitivity and reward, and use of stimulants, and were very low in sensitivity of punishment. So what do we do to them? We punish them, right? Very low in anxiety sensitivity and very low in hopelessness.

- 6.50** We did a follow-up study. We followed these people for three years and did an extensive clinical interview, the presence of current or remitted alcohol and substance use disorder. Of 23 subjects interviewed, we interviewed 11. Others had moved on. 50% of them were sensation seekers. Then we basically did a diagnosis, and we found that half of the ones we interviewed were now legally diagnosable as having a drug problem or an alcohol use problem.
- 6.51** What it showed, and this was for that first measure of faces, it was the anxiety-sensitive individuals who we were able to follow, who showed the most discrete response, the most dampening to the excitatory stimulus of the fearful faces. They're the ones who later became addicted. So we followed subjects on that path from not a problem, but a likelihood, to the fact that they are diagnosable as having a problem.
- 6.52** And these four graphs point out basically the top line, what I was just saying. The bottom line looks at transitioners in terms of those who were the individuals who were not anxiety-sensitive but responded to the mass stress. And again, they were the ones who showed the greater effect, who were likely to transition.
- 6.53** We're taught to use drugs. We live in an environment where pills are pushed on us everywhere. We define problems in terms of requiring medical solutions. This is a graph that shows the estimated rates of incidence of chronic pain among adults. You can see that, for a large percentage of individuals, at least some chronic pain is a natural concomitant of life. So why does the pharma industry exist? To make money. Each year, they spend 6 to 8 billion targeting consumers. Only the US and New Zealand allow direct, that I know of, direct consumer ads. And they're going to do what they do to make that money.
- 6.54** I've given a lot of talks to physicians paid by pharmaceutical companies. And they're great gigs because they're always at really nice hotels. There's all kinds of neat stuff that goes on, and they're really well taken care of, and it's really rewarding. And the people who are brought there, the MDs, they have to listen to you for a little while, but most of the time, everything else is planned and set up. The whole point is, it's sold as an educational experience. But it's really, it was for a while a family outing paid for by drug companies.
- 6.55** How do physicians learn about drugs? From drug salesmen. I know a drug company that will only hire really attractive women to be drug salesmen. They do that on purpose. So we shouldn't get upset; that's their goal. That's what they do. They need to be monitored. So it's easy to criticize the industry because we want them to be pure of heart, but that is naive because no one's pure of heart.
- 6.56** They spend more on marketing than research and development. Maybe that'll change now that they can use AI to develop all their new drugs. They create diseases where they don't exist. And one way they do that is, they get physicians interested in a problem that looks like it could have separate identity. And so why do we see this

growth in classification systems? I mean, how did we get from 100 and some disorders in 1952 to 500 and some today in mental illness?

- 6.57** Some of it's very legitimate, and the definitions are being more precise. But many are advocated and politicized and pushed into acceptance, which creates a whole new market. Direct-to-consumer advertising can be misleading. Direct-to-others can have half-truths. And so you can easily develop a conflict of interest.
- 6.58** Furthermore, there's studies that have been done that show that 60% of the individuals sitting in a GP's office don't belong there. They're not ill. Of course, now the majority of times you go to the GP, it's for a checkup. But if people come with a problem, isn't there a contract that's involved? I come to you with a question, something's wrong with me, what are you going to do? If there are a lot of people there who don't really have a problem, what do you do?
- 6.59** You give them a drug. A placebo would be the best, but that's illegal. So it's really an antidepressant. And that's why there are so many antidepressants that are written as prescriptions. And we know there's higher prescription rates in physicians who are younger and on a personality level, more authoritarian. "Here, take this, get out" kind of thing.
- 6.60** And that may be the case why women are prescribed drugs two to three times more than men. It's a way in which, the woman might just want to talk, but the physician doesn't have time. It doesn't matter whether they're male or female. They just don't have time because somebody put a revolving door on their door, and they have to have you in and out on a time limit basis because they don't work for themselves anymore.
- 6.61** We did a study where we looked at 180 women randomly picked out of voter lists. We were able to divide them into three groups, a group that were taking anti-anxiety drugs. And those that were, of that, we had three groups. Regular users, irregular users, and non-users. What we did with them is we did a problem-solving test where we asked them to listen to tape recordings of scenarios, and they were interactions that we had concocted between a woman and somebody was listening.
- 6.62** All that we required our subjects was to answer these four questions: What's the problem? What's being discussed? What do you think could have led up to this situation? How would this problem end, and what is the solution? And what alternative solutions could you see for such a problem? That's a standard way in which you can assess problem solving in subjects. The results were interesting.
- 6.63** In terms of the ability to come up with a definition, people who weren't taking any drugs did really well. In fact, they did well on all four of those questions. The regular users and, to a lesser degree, the irregular users, did significantly poor. That is, to begin with, they were listening, and they couldn't even formulate the problem. Then obviously they couldn't make up the right solutions. It was all significantly different.

- 6.64 These are the people who are on scripts, that have gone to a physician; and what we're hypothesizing backwards is that they went and complained about something, and it wasn't clear what they were complaining about, and they were given a script. But it's their cognitive style that's creating the issue, a fundamental deficit in problem solving. The ability to identify with a formal system, with overidentification with peer groups, are also factors that appear to increase drug use risk.
- 6.65 For adolescents, here's a list of risk factors: low grade point, lack of religiosity, early alcohol use, low self-esteem, psychopathology, poor relationship with parents, lack of social conformity, sensation-seeking, perceived peer drug use, perceived adult drug use. You get a graph like this in terms of those number of risk factors. As the number of risk factors grows, so does the frequency in substance abuse. You can see it grows rather strongly as the number of risk factors increase.
- 6.66 To a certain degree, we learn to take drugs, legal or illegal. They can make us feel good, some more than others. It's only because there are negative factors, attached fear, personal, societal, that we don't. Cultural factors and general effect. What more than anything else has decreased smoking? Yeah, exactly. Money. Price. Price has risen so that people just said, "That's it. I'm going to live with my pain, whatever it is." And that's why it takes a number of chances to give up, because you have to get really ornery over time before you finally throw it in completely.
- 6.67 But we should be looking at some of these drugs. Instead of dealing with situations where there aren't costs, there might be other ways to think of costs that go beyond money. How we think certain drugs act in general, get facilitory, particularly with young kids. I did a study where we looked at cannabis effects in Montreal versus California. In Montreal, it was a spiritual, mystical, sexual cohesion thing. And in California, it involved thought disruptions and social withdrawal. At that time, because the dosage was low, it was really an all-purpose thing.
- 6.68 And to illustrate that, I had a lab that was made up like a pad, curtains hanging from the ceiling and all kinds of pillows and massive stuff that people could sit on. I got thousands of people stoned. What was fascinating is, first of all, men would get completely stoned differently than women. Totally different. Take a group of male strangers, put them in a room, get them stoned, and they just sit there, stoned. Get a bunch of women in there, and the noise level would knock you over. Just chatter away, having a great time. Totally different responses.
- 6.69 And then you'd ask them to make certain judgments. So you take them out in the hall and say, "I want you to tell me, how far is it from here to there?" And we have a place marked off. And they'd say, "Hey, man, that's forever." No, no, no. "Here's five bucks. How far is it from here to there?" Bam, right on, they'd have it exactly. The whole response at that time was pretty much of a societal, directed, social thing. It was really the world's most used placebo.

- 6.70** I did want to say a little bit about mom because mom teaches us to use drugs, and it's natural. You have a child that's disturbed or something's wrong, and you don't know what to do; well, we've learned to use pharmaceuticals. So at a basic level, we've all learned that's what drugs are for. And we're taught that as kids. Hopefully, we don't go to that level of, if you've got any kind of a problem, that's the way you solve it. But unfortunately, we learn very quickly and often don't learn to discriminate.
- 6.71** So let's talk a bit about treatment. Those who are treated in their lifetime for addiction is very low in number. It's only about 8% for abuse and 38% for dependence. Deference is often given to treating comorbid disorders. If you see a physician and you're addicted to alcohol but you're depressed, the treatment you will get is for depression. It won't be for alcohol.
- 6.72** Relapse is also the rule. It is a case that generalized prevention programs have been shown to increase drug use rather than decrease it. There was a series of studies where the police would go and talk to kids in schools. I can't remember the name of it. [inaudible student] Yeah. It's a good study that shows that DARE created more drug use. Basically, the message somehow got to the kids, "Hey, that's something to try," even if you would have a big cop in his uniform and a gun telling you how dangerous this is.
- 6.73** It's important to get the kids because early drinking is highly predictive about who's going to be dependent as they get older. So you want to create a negative environment, not a positive environment, for alcohol. Athletes selling booze probably is not a good thing. The problem we have and why a person, a psychiatrist, is apt to treat depression rather than alcoholism is, we don't have really good treatments. I mean, part of the reason why we have harm avoidance is out of frustration, is out of not knowing what's really going to work and what to do.
- 6.74** The majority of these drug programs, if they're working, are basically replacing the drug that the street person is using with another drug. And it's another dopaminergic drug that's working. But going beyond that, we're not too great. Nevertheless, there are some behavioral approaches that have some positive effect sizes. Contingency management, cognitive behavior therapy, cognitive behavior therapy with contingency management, and relapse prevention. The same is true of other substance uses in terms of treating polysubstances, cocaine, opioids, and cannabis. But I don't believe those affect size.
- 6.75** What we really know is about 80% are never treated. That doesn't matter whatever the treatment is. Here's a treatment for alcohol use disorder using common drugs that are used—acamprosate, naltrexone, and disulfiram. Basically, compared to placebo, the numbers required to prevent one person from returning to drinking meant that you had to treat 11 others for acamprosate and 18 others for naltrexone. They did the experiment wrong. If they had challenged these guys and tested them to see if they

could reduce it with naltrexone, they could have tested the drug effect in a challenge, rather than in weeks of treatment.

- 6.76** Again, what we're starting to see is the use of these procedures that are atypical and, for practical purposes, illegal. Here is psilocybin being given to individuals who have an alcohol use disorder with significant and positive findings. That data is, again, starting to accumulate at a number of areas. It wasn't just depression. It's really across the board of mental illness.
- 6.77** What this figure shows is that you can intervene at a number of places along the addiction dimension. You can intervene at predisposing factors. It's probably where you should intervene, at adolescent developmental factors, during drug maintenance or during addiction. What I'm going to do is show you that this is a report from that group that tells you whether it's worthwhile doing something. They basically say here that, no, there aren't any good measures or interventions to prevent drug use, to which I'm going to say they're wrong.
- 6.78** To facilitate treatment, you need a refined population, answers to what and in whom questions, and a delineation of mechanisms. This is a study that Trish Conrod did. She was a student of mine and wrote a grant. We looked at, basically, it was 300 substance-abusing women. We advertised: "If you have a substance use problem, and want to learn more about yourself, come to this address at this time, or call this number and we'll make an appointment." We got overwhelmed. We had to pull the ad after one or two days. We said 300 people is enough.
- 6.79** We said we're going to spend the morning with them. We brought people in and had this idea that maybe drug abuse and comorbid pathology have similar etiology. That is, what the drug does to an individual might be important relative to their other mental illness. And that's what we're going to ascertain. We gave them an extensive battery of tests, and that's what's on the left there. It's about any, every measurement you could come up with that had any kind of validity and reliability.
- 6.80** Then what that is, is a factor analysis of the results of these 300 women. You could see there was an anxiety-sensitive factor. There was a sensation-seeking factor. There's an impulsivity-hostility factor, and there's an introversion-hopelessness factor. We did a cluster analysis to see how these various factors related to each other. What the cluster analysis shows is that there are five clusters. One, the hopelessness, which is basically a depression factor. These individuals are very high in hopelessness and very low in sensations and impulsivity.
- 6.81** There was a fearful factor. They are high in fearfulness and low in everything else. There was a sensation-seeking factor. High in sensation seeking, low in everything else. And there was an impulsive factor, where they were high in sensation seeking and impulsivity. It shows you, first of all, that impulsivity, that sensation figures in two of

these clusters. Then there was a low personality risk factor. But there were very few women that fit that. They all had a significant other who was addicted.

- 6.82** By the way, all of these women were diagnosable with an addiction. This is where we looked at the diagnosis relative to which of those factors they fell into. What we found was anxiety-sensitive women were two times more likely to be diagnosed with simple phobia than the others; almost six times more likely to be diagnosed with somatization; and three times more likely to be abusing anxiolytic medication.
- 6.83** The introverted hopeless were more likely to be depressed, three times more likely to be major depressive episode, twice as likely to be diagnosed with a panic disorder, and almost three times more likely to show social phobia. They were also almost three times more likely to be opioid dependent. The sensation seekers didn't have any particular pathology, but were four times more likely to be alcohol dependent, and the impulsives were three times more likely to be antisocial, and twice as likely, almost three times, alcohol dependent and almost two times, cocaine dependent.
- 6.84** The kind of drugs that people are using is fitting their kind of pathology. We did a brief intervention where we created a motivationally matched training program that lasted a half hour. And so we had three groups. We had individuals who watched an educational film, same time, about drug problems. We had a group where, if they were in the depressed category, we taught them how not to be anxious. So we did the wrong thing.
- 6.85** We had a motivationally matched group where what we did is we, if they were depressed, we gave them a brief intervention on how not to be depressed. But overall, what we found is, over six months, half of the women in the motivationally matched group were not dependent anymore. That's all we did, 30 minutes. But maybe it was the whole day where they learned about themselves that really made an effect.
- 6.86** Out of this, we developed an instrument called the SURPS, Substance Use Risk Profile, which measures a person on this dimension of hopelessness, anxiety, sensitivity, impulsivity, and sensation seeking, which has caught on around the world, actually. It has been used a lot, and we're using it in some intervention programs. Because this is the way things are seen.
- 6.87** We see people—we're able to categorize people now who have drug problems that speaks to a different intervention for each. We look at people who are impulsive, abuse stimulants, have externalizing problems, for sensation-seekers, who also fit that same dimension. Then we have people who fall in the hopelessness and anxiety sensitivity category. They require a totally different kind of intervention. We're differentiating the populations.
- 6.88** This is the study we do and have been doing, where we go into a school, teach teachers or counselors or anybody who wants to do it. I want to intervene very briefly with kids in groups. But it's after we've assessed them and can put them in one of these

categories. We have kids in control groups, too. We look at thousands of kids in various school systems. The program is being used around the world.

- 6.89** We have a number of studies that show that we take these kids who are high in negative thinking, high in anxiety sensitivity, high in impulsivity, high in sensation-seeking, do follow-ups that last for months, and can show you curves where their usage of drugs, their increase is lower after that intervention than if they didn't have it. It doesn't necessarily reduce things to zero or stop it, but you decrease the number of kids who get involved.
- 6.90** Here's another graph where there's a marked depression in illicit drug use as a function of going through the program, through to 24 months. Survival rate, the survival curve, showing the same result. A dramatic effect on stopping binge drinking. Look what happened to their depression scores as a function of going through the program. Even side effects you don't expect. Here's one graph that shows how panic attacks were reduced. And another one, how we reduced shoplifting. This was in a British sample.
- 6.91** This is just another study where we got significant changes in depression, suicidal ideation, anxiety, and conduct problems, with very brief interventions. What I'm advocating for is to attend to risk and to try to prevent. In other words, at that first stage, to operate at a school, family level, to diagnose the etiology of a specific problem. That's essential because each of the interventions that we've done is targeted. If you're anxiety-sensitive, that's what we target. If you're depressed, that's what we target.
- 6.92** We have workbooks and all kinds of shtick that one can use to implement that. You have to diagnose the etiology of specific problems to know what you're dealing with. But that brief scale, the SURPS, will tell you where a person fits. Then you have to treat the physical and psychological effects of a drug or behavior if they've occurred, and that's much more difficult. Prevention is far more effective, and at least reducing the amount of individuals who get involved.
- 6.93** Finally, we have to strive for some societal consensus. If you've been following our discussions, it's clear that one of the present problems is, societally, we don't have much of any consensus. In fact, we are becoming separate societies—and separate societies in the name of symmetry, which is mind-boggling. It just doesn't work.
- 6.94** Here's a test for you. If you can start out the day without caffeine, if you can always be cheerful, ignoring aches and pains, if you can resist complaining and boring people with your troubles, if you can eat the same food every day and be grateful for it, if you can understand when your loved ones are too busy to give you the time of day, and if you can take criticism and blame without resentment, if you can conquer tension without medical help, if you can relax without alcohol, and if you can sleep without the aid of drugs, then you're a dog. Bark on. Thanks.